UNIVERSIDADE DA CORUÑA

Rnasa Imedir

Unit 1.4

Decision Trees

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Machine Learning I

fic facultade de informática da coruña

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Decision Trees

- Decision tree:
  - It is a representation of the decision processes involved in classification tasks
  - Items:
    - Leaves: describe the label associated with a classification
    - Nodes: describe the question about a certain attribute
  - Branches of a node:
    - They represent the different values that the attribute of the node can take

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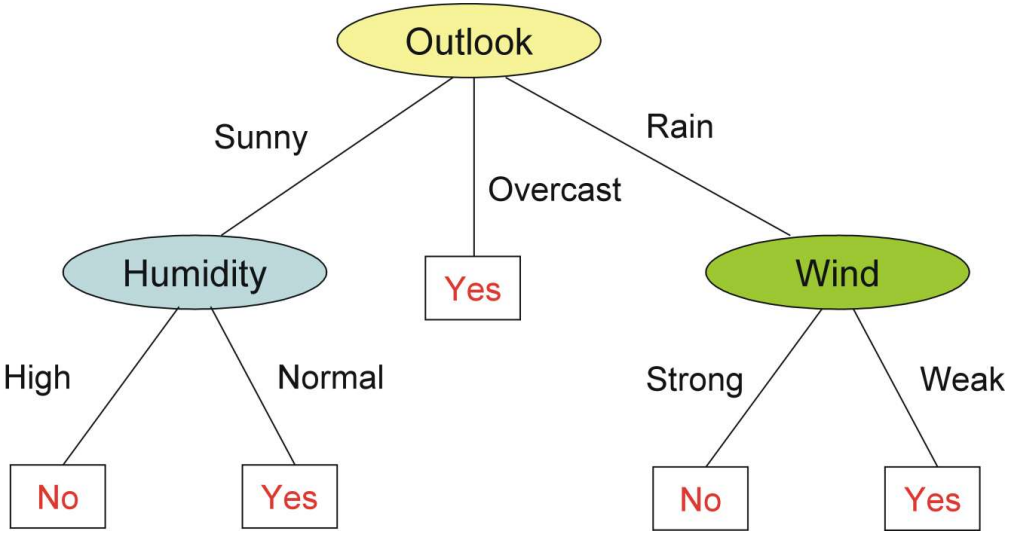
Decision Trees

- Example:
  - Tree to classify whether a day is suitable for playing tennis:



- Attributes: Outlook, Humidity, Wind
- Classes: Yes / No

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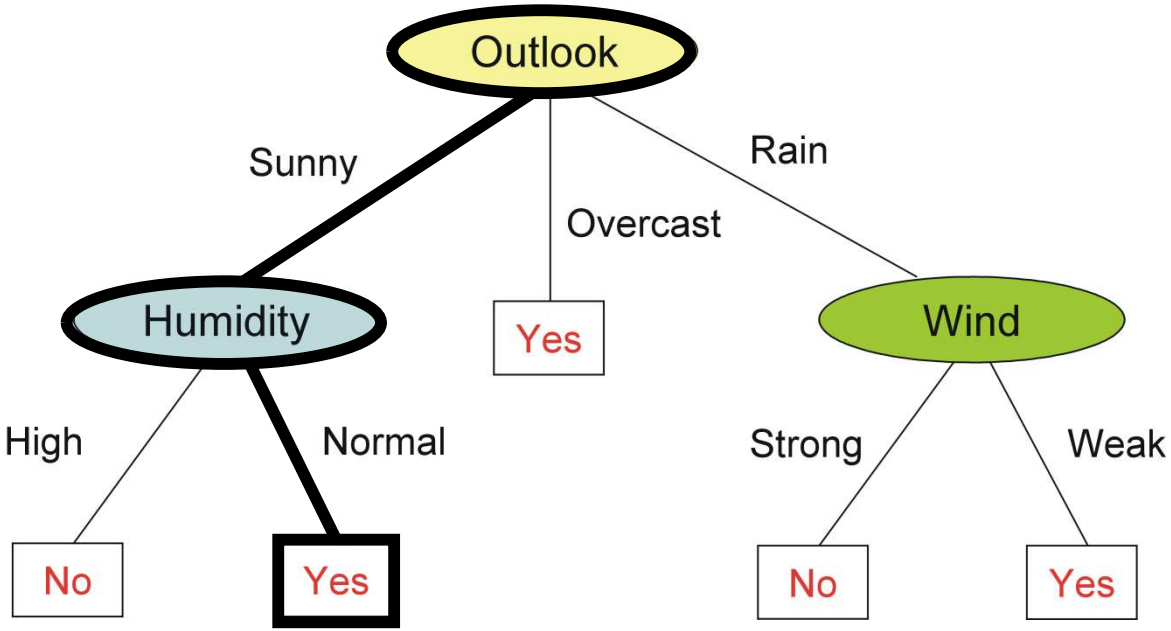
Decision Trees

- Example:
  - Tree to classify whether a day is suitable for playing tennis
    - If a new instance appears, the tree is traversed
      - Example: (Outlook = Sunny, Temperature = Hot, Wind = Strong, Humidity = Normal)

→ classified as *Yes*



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Decision Trees

- In the case of classifying into two classes, the tree represents a disjunction of conjunctions of constraints on the attribute values of the instances
  - One path = conjunction of attribute tests
  - The whole tree = disjunction of these conjunctions
  - Example: previous tree:
    - (Outlook = Sunny ∧ Humidity = Normal) v (Outlook = Overcast) v (Outlook = Rain ∧ Wind = Weak)

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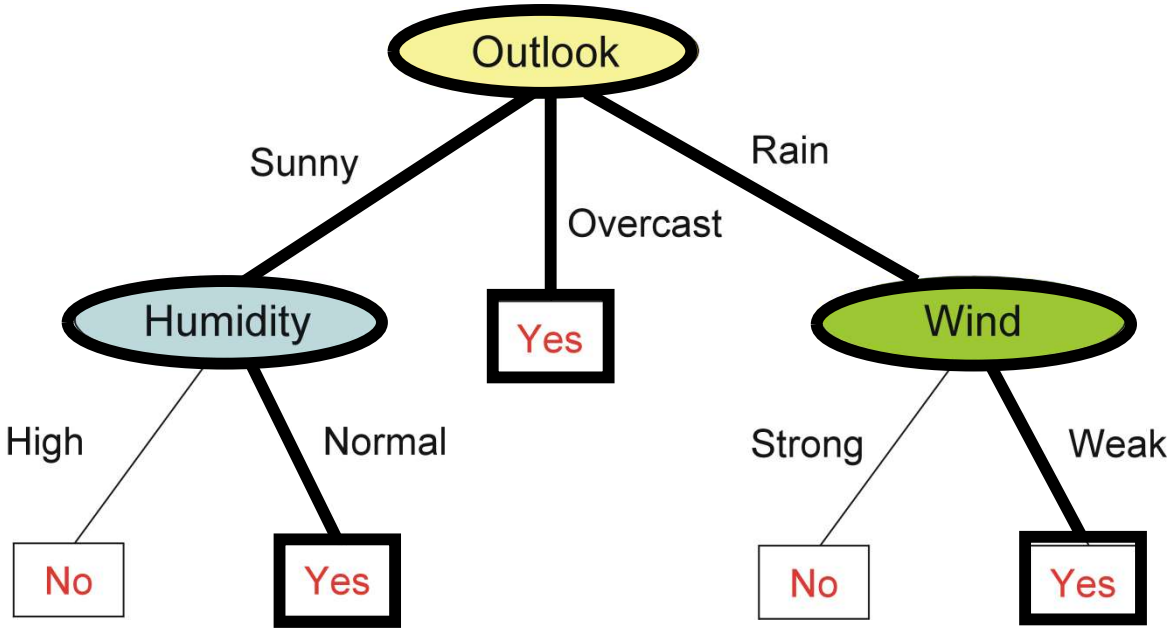
Decision Trees

- Disjunction of conjunctions:
  - Example: previous tree:

$$(Outlook = Sunny \land Humidity = Normal) \lor$$
$$(Outlook = Overcast) \lor$$
$$(Outlook = Rain \land Wind = Weak)$$



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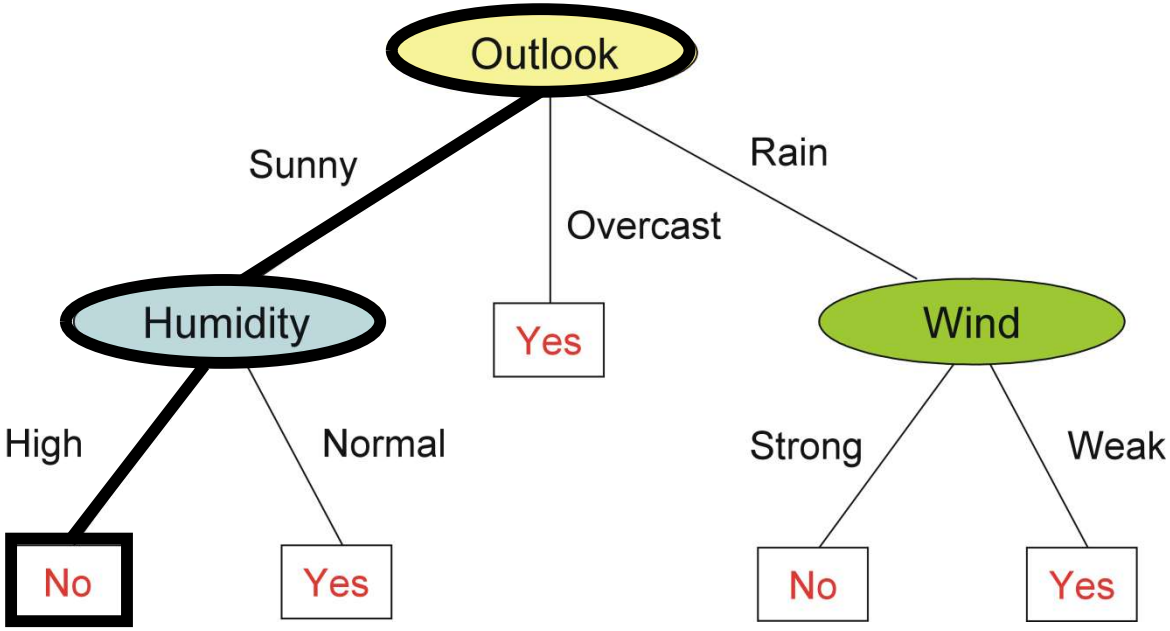
Decision Trees

- In addition, a series of rules can be generated from the tree:



- A1: If (Outlook = Sunny) AND (Humidity = High) then PlayTennis = No

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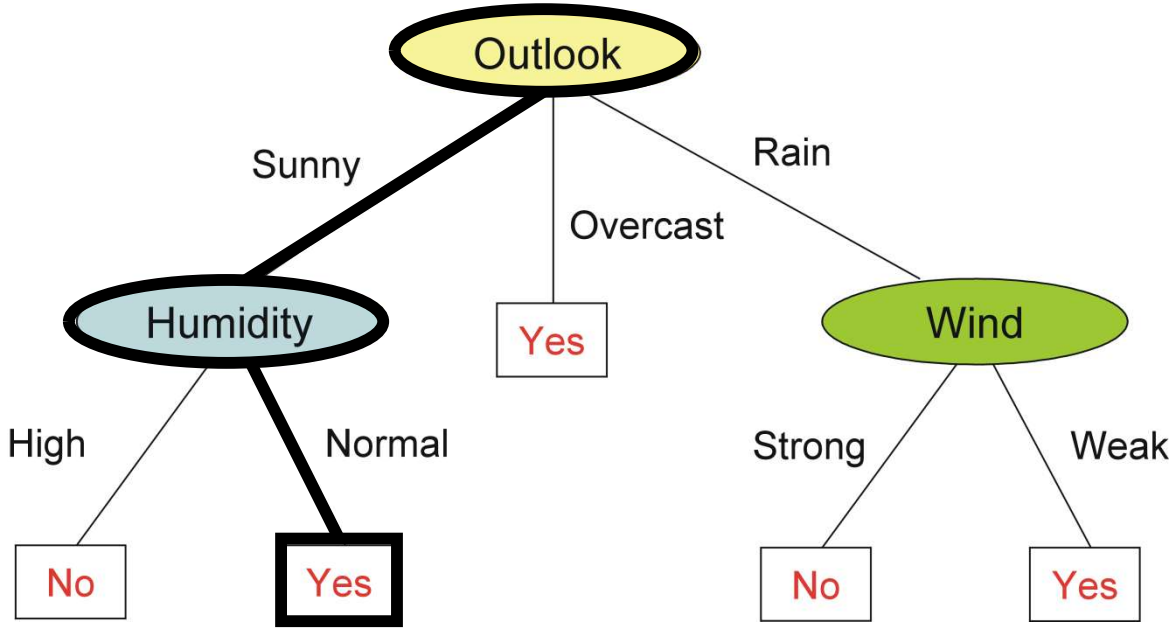
Decision Trees

- In addition, a series of rules can be generated from the tree:



- A1: If (Outlook = Sunny) AND (Humidity = High) then PlayTennis = No
- A2: If (Outlook = Sunny) AND (Humidity = Normal) then PlayTennis = Yes

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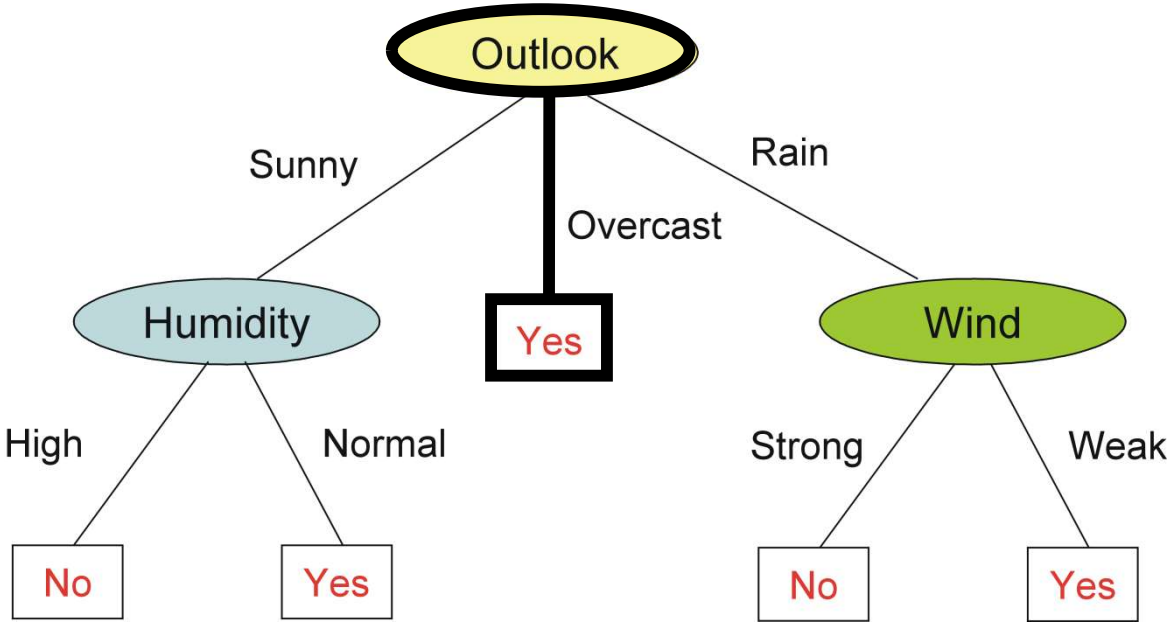
Decision Trees

- In addition, a series of rules can be generated from the tree:



- A1: If (Outlook = Sunny) AND (Humidity = High) then PlayTennis = No
- A2: If (Outlook = Sunny) AND (Humidity = Normal) then PlayTennis = Yes
- A3: If (Outlook = Overcast) then PlayTennis = Yes

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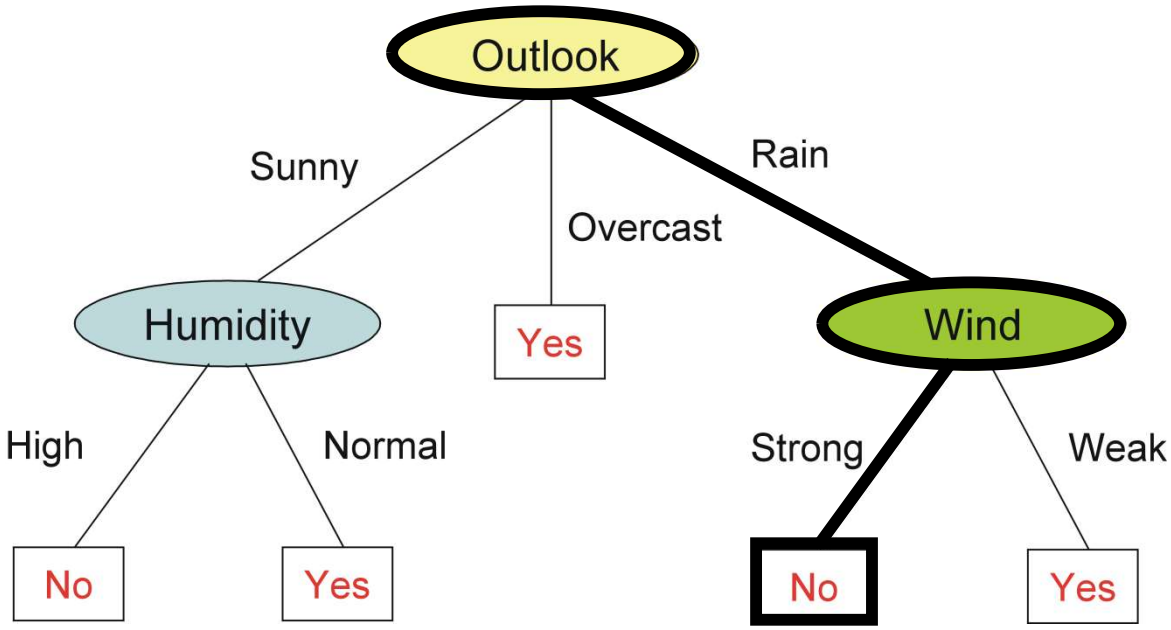
Decision Trees

- In addition, a series of rules can be generated from the tree:



- A1: If (Outlook = Sunny) AND (Humidity = High) then PlayTennis = No
- A2: If (Outlook = Sunny) AND (Humidity = Normal) then PlayTennis = Yes
- A3: If (Outlook = Overcast) then PlayTennis = Yes
- A4: If (Outlook = Rain) AND (Wind = Strong) then PlayTennis = No

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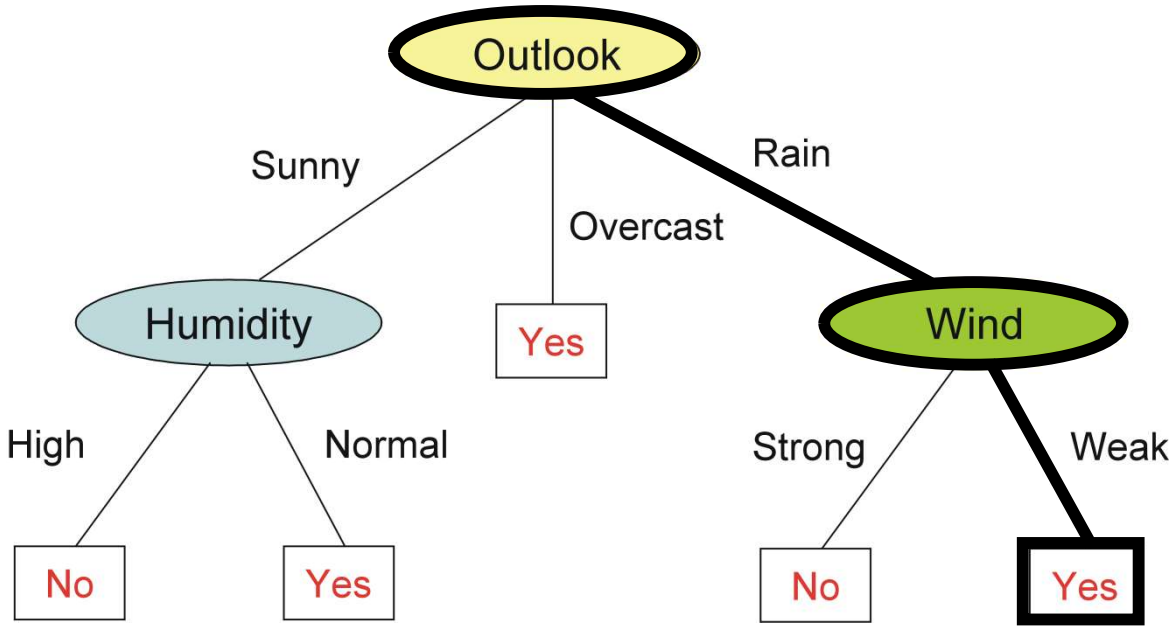
Decision Trees

- In addition, a series of rules can be generated from the tree:



- A1: If (Outlook = Sunny) AND (Humidity = High) then PlayTennis = No
- A2: If (Outlook = Sunny) AND (Humidity = Normal) then PlayTennis = Yes
- A3: If (Outlook = Overcast) then PlayTennis = Yes
- A4: If (Outlook = Rain) AND (Wind = Strong) then PlayTennis = No
- A5: If (Outlook = Rain) AND (Wind = Weak) then PlayTennis = Yes

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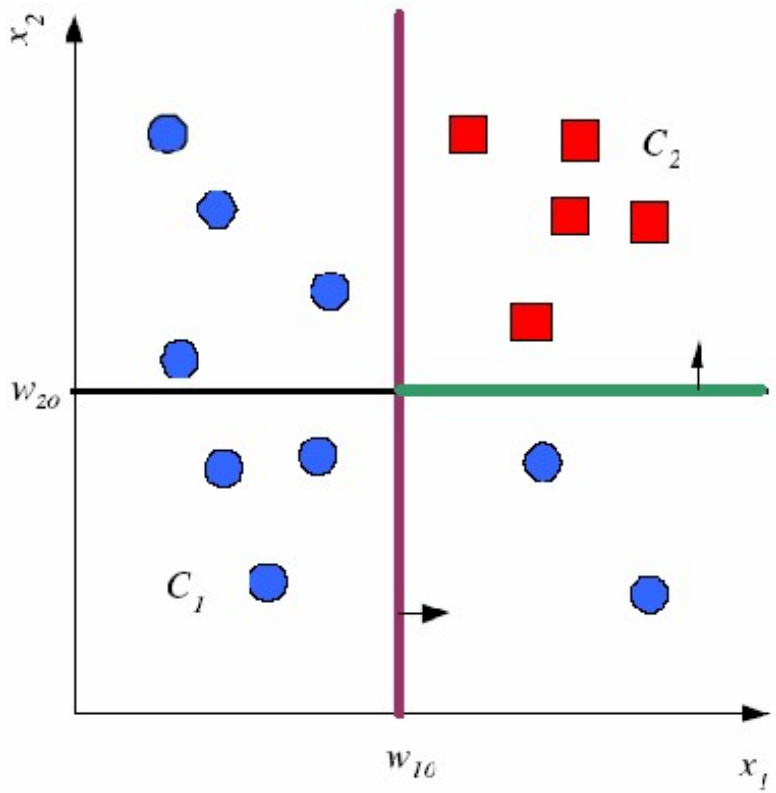
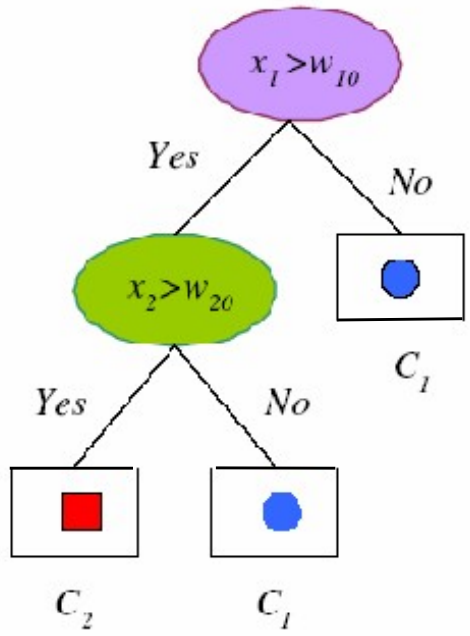
Decision Trees

- Division of the space:
  - The space is divided into regions labeled with a single class and are hyperrectangles





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Decision Trees

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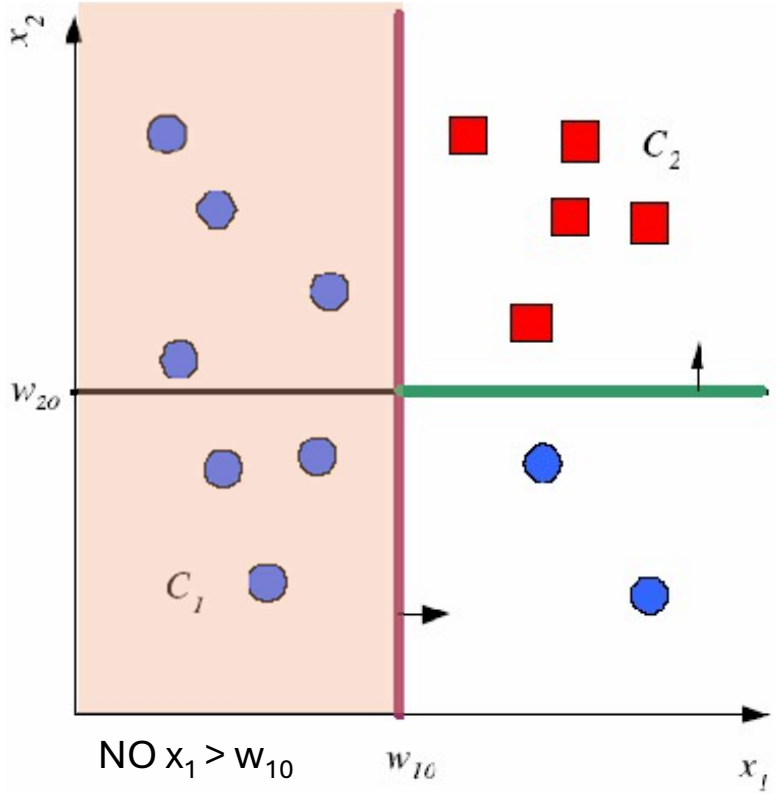

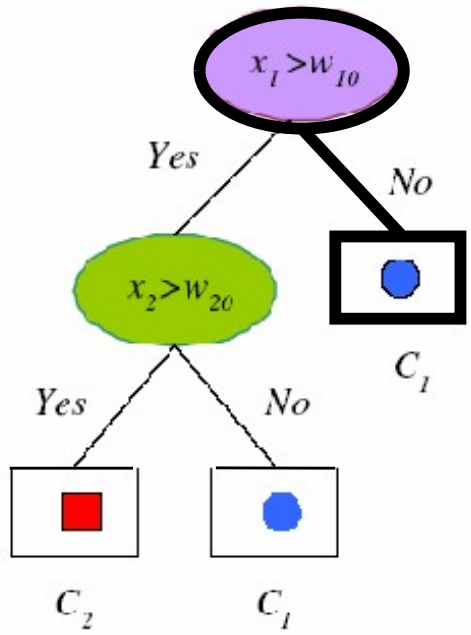

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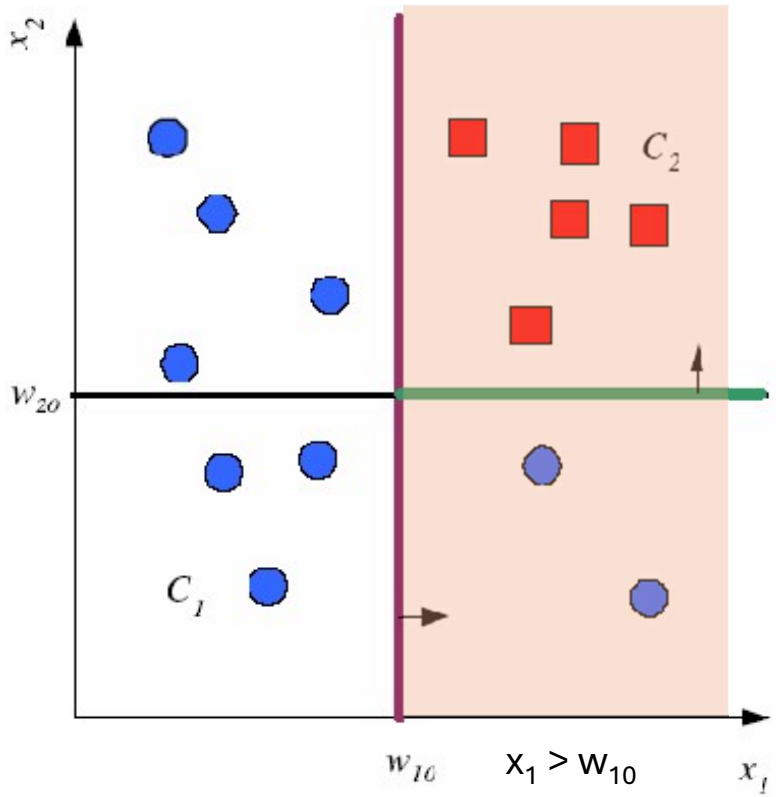

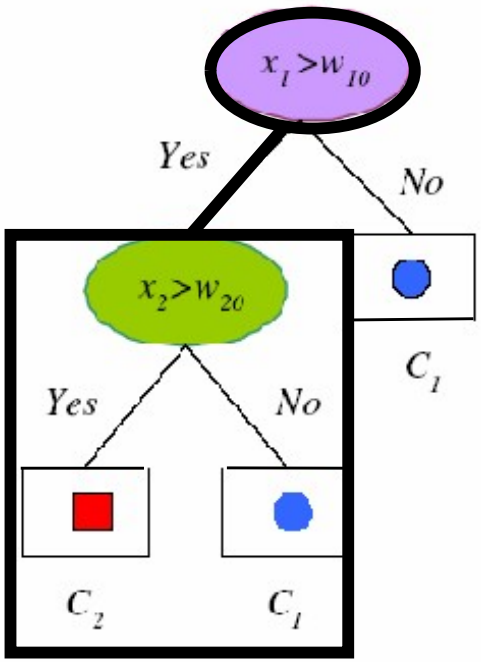

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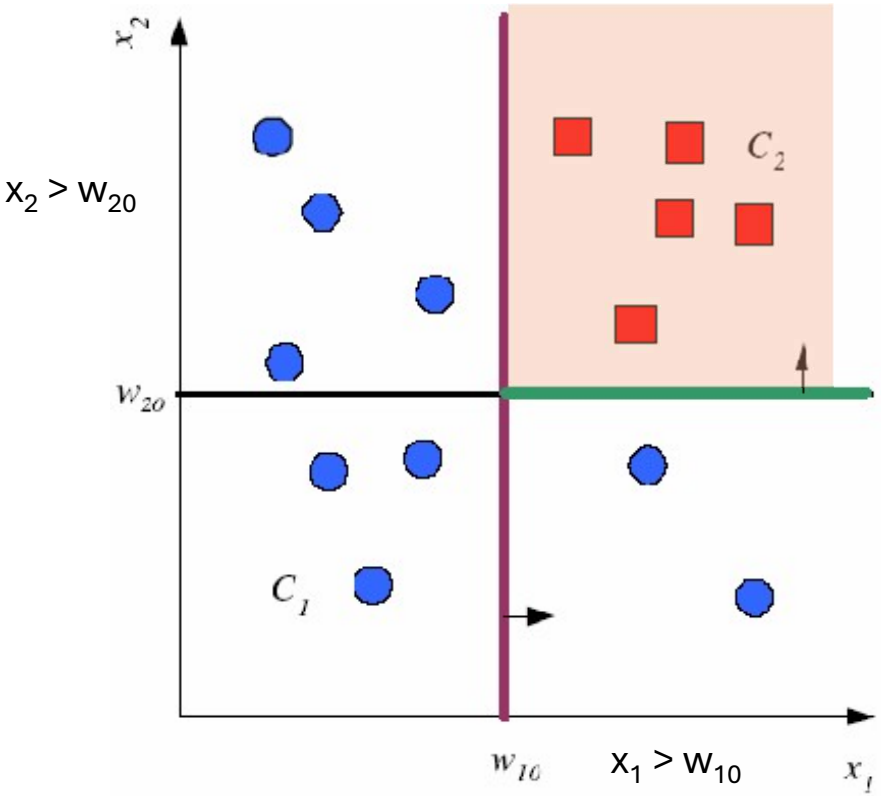

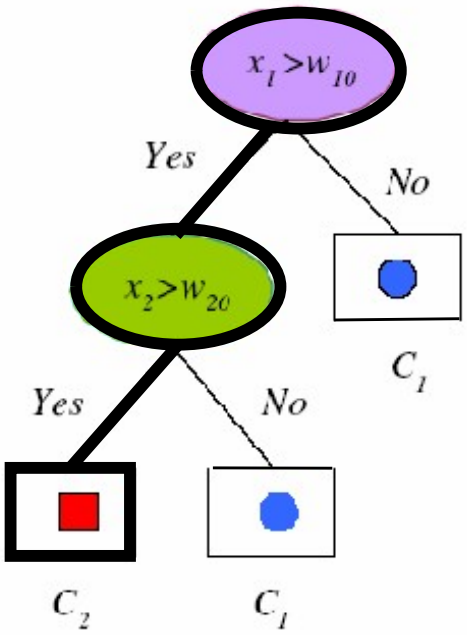

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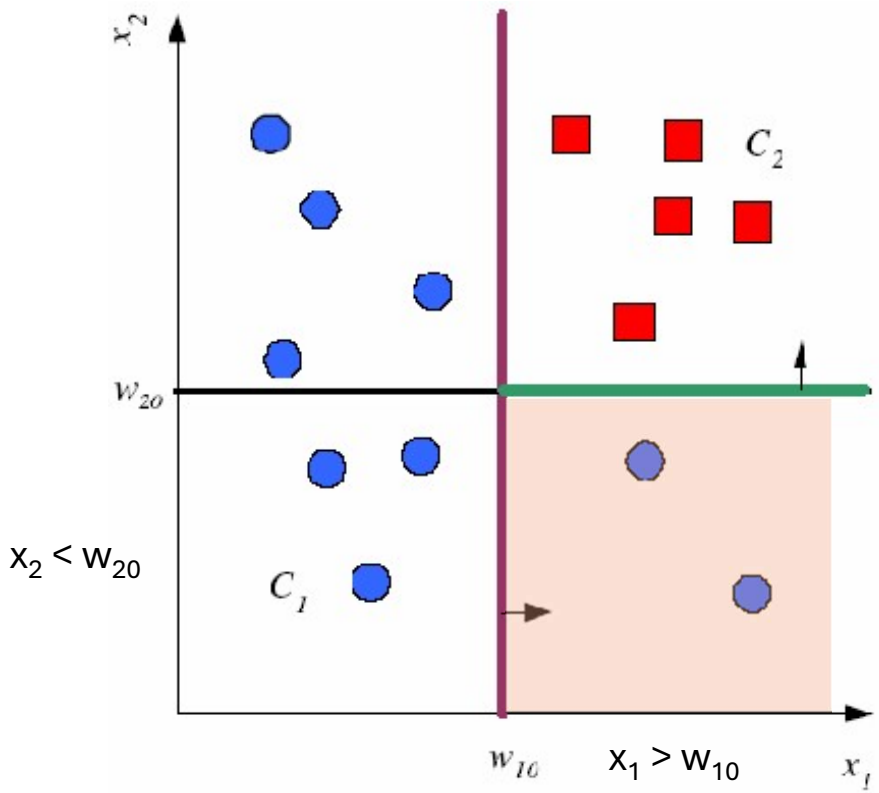

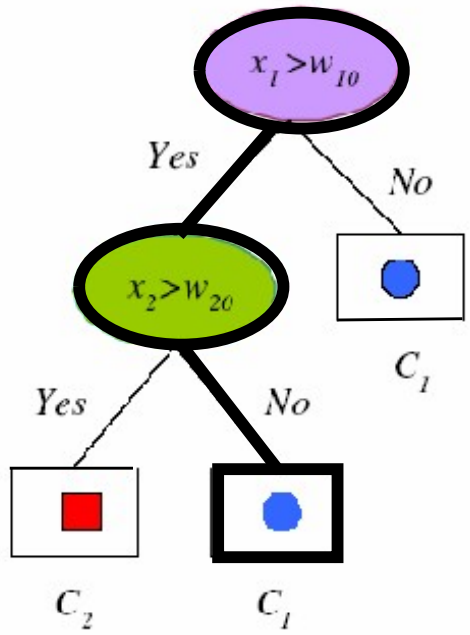

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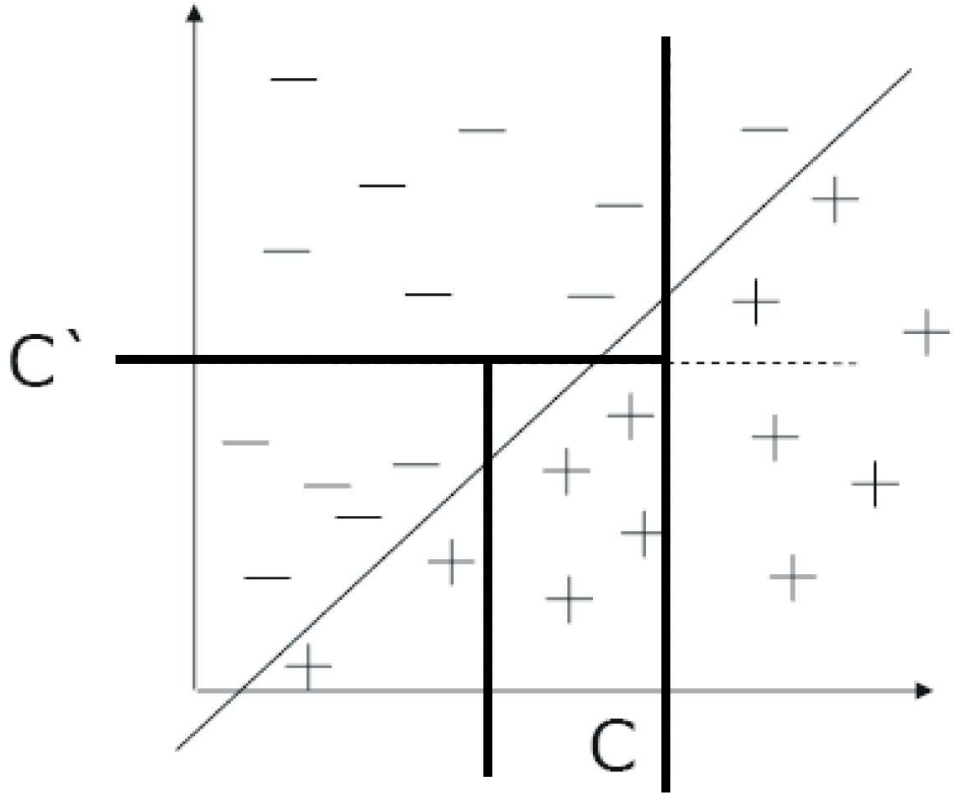

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Decision Trees

- Tree generation:
  - Several algorithms available
  - Objectives:
    - Given a set of classified examples (supervised learning), generate the optimal decision tree
      - The one that allows the classification to be described with the least number of questions possible
  - Main algorithms:
    - ID3
    - C4.5
    - CART
    - MARS

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Decision Trees

- ID3 (Iterative Dichotomiser 3) (Quinlan, 1986)
  - Based on the CLS (Concept Learning Systems) algorithm which used only binary attributes
  - Build the tree from top to bottom:
    - Which attribute to select as root node?
      - Each attribute is evaluated to determine how well it classifies the examples on its own.
    - The best attribute is selected as the node and
      - The tree is opened for each possible value of the attribute
      - The examples are placed in the branches
        - Based on the values of that attribute selected as a node
        - "Split" the dataset into subsets, one for each branch

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Decision Trees

- ID3 (Iterative Dichotomiser 3) (Quinlan, 1986)
  - Based on the CLS algorithm (Concept Learning Systems) which used only binary attributes
  - Build the tree from top to bottom:
    - Repeat this process of selecting attribute / opening tree using the examples associated with the branch we are on
      - Always forward, searching among the unused attributes on this path
    - Stop when the tree correctly classifies the examples or when all attributes have been used
    - Label the leaf node with the class of the examples

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Decision Trees

- ID3 algorithm:
  - ID3 (instances)
    - IF all instances are of the same class C THEN return Leaf(C)
    - ELSE IF the instance set is empty THEN return Leaf(Default_Class)
    - ELSE IF the instances set does not contain any attributes THEN return Leaf(Majority_Class)
    - OTHERWISE
      - Choose attribute A with highest information gain
      - Create node with that selected attribute
      - For each value V of attribute A
        - Create a branch with the value V
        - Select instances with value V of attribute A
        - Remove attribute A from this set of instances Cv
        - Assign to the branch the tree returned by the recursive call ID3(Cv)
      - Return node

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Decision Trees

- ID3 (Iterative Dichotomiser 3) (Quinlan, 1986)
  - Key step: how to select the attribute to test at each node of the tree
    - Select the most useful attribute
      - The one that best separates the examples
    - The amount of mutual information is used as a measure of the value of each attribute
      - The attribute with the highest information gain is chosen
      - It is expected to reduce the entropy (uncertainty), caused by partitioning the examples according to this attribute
  - Definitions:
    - C: set of classes
    - n: number of data samples
    - n_c: number of instances of class $c \in C$

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - $C = \{\text{Normal, Carcinogenic}\}$
      - $n = 6$

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - C = {Normal, Carcinogenic}
      - n = 6
      - $n_1 = 3 \rightarrow$ 3 instances of class 1

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - C = {Normal, Carcinogenic}
      - instances of class 2

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - A: attribute set
  - A_i: attribute i of the set of attributes A
  - V_i : set of values of A_i

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - A = {Antennas, Tails, Nuclei, Body}
      - A_1 = Antennas A_2 = Tails A_3 = Nuclei A_4 = Body
      - $V_1 = \{0,1,2\}$ $V_2 = \{0,1,2\}$ $V_3 = \{0,1,2\}$ $V_4 = \{White, Striped\}$

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - n_{ij}: number of instances that in the attribute A_i take the value $j \in V_i$

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - A = {Antennas, Tails, Nuclei, Body}
      - A_4 = Body, V_4 = {White, Striped}
        - $N_{41} = 1$

Instances that in attribute 4 'Body' take the value 1 'White'

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - A = {Antennas, Tails, Nuclei, Body}
      - A_4 = Body, V_4 = {White, Striped}
        - $N_{42} = 5$

Instances that in attribute 4 'Body' take the value 2 'Striped'

Example	Antennas	Tails	Nucleus	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - n_{ijc}: number of instances that in the attribute A_i Ai take the value j and belong to class c

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - A = {Antennas, Tails, Nuclei, Body}
      - A_4 = Body, V_4 = {White, Striped}
        - $N_{421} = 3$

Instances that in attribute 4 'Body' take the value 2 'Striped' and belong to class 1 'Normal'

Example	Antennas	Tails	Nucleus	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Example: cell dataset:
    - A = {Antennas, Tails, Nuclei, Body}
      - A_4 = Body, V_4 = {White, Striped}
        - $N_{422} = 2$

Instances that in attribute 4 'Body' take the value 2 'Striped' and belong to class 2 'Cancer'

Example	Antennas	Tails	Nucleus	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node ("Body"):

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

The dataset is split:

Body

White

Example	Antennas	Tails	Nuclei	Class
2	1	0	1	carcinogenic

Striped

Example	Antennas	Tails	Nuclei	Class
1	1	0	2	Normal
3	1	2	0	Normal
4	0	2	1	Normal
5	1	1	1	carcinogenic
6	2	2	1	carcinogenic

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node ("Body"):
    - For each branch:
      - Branch "White": 1 instance
        - All instances are of the same class: total certainty
          - **Entropy 0**

Example	Antennas	Tails	Nuclei	Class
2	1	0	1	carcinogenic

      - Branch "Striped": 5 instances
        - 3 "Normal" class instances and 2 "Carcinogenic" class instances
          - There is no certainty
            - **Entropy > 0**

              (entropy = 0.971)

Example	Antennas	Tails	Nuclei	Class
1	1	0	2	Normal
3	1	2	0	Normal
4	0	2	1	Normal
5	1	1	1	carcinogenic
6	2	2	1	carcinogenic

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node (“Body”):
    - Therefore, the entropy as a result of this division with the attribute "Body" is
      - Weighted sum of the entropies of the branches

$$\frac{1 \cdot 0 + 5 \cdot 0.971}{1 + 5} = 0.81$$

- gives an measure of the uncertainty in the branches when dividing with that attribute

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node ("Queues"):

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

Tails

Example	Antennas	Nuclei	Body	Class
5	1	1	Striped	carcinogenic

Example	Antennas	Nuclei	Body	Class
1	1	2	Striped	Normal
2	1	1	White	carcinogenic

Example	Antennas	Nuclei	Body	Class
3	1	0	Striped	Normal
4	0	1	Striped	Normal
6	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node ("Tails"):
    - For each branch:
      - Branch "0": 2 instances
        - Equal number of instances of each class: total uncertainty
          - **Entropy 1**
      - Branch "1": 1 instances
        - All instances of the same class: full certainty
          - **Entropy 0**
      - Branch "2": 3 instances
          - 2 instances of the "Normal" class, 1 "Carcinogenic"
            - **Entropy > 0**
              - (entropy = 0.9183)

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node ("Tails"):
    - Therefore, the entropy as a result of this division with the attribute "Tails" is
      - Weighted sum of the entropies of the branches

$$\frac{2 \cdot 1 + 1 \cdot 0 + 3 \cdot 0.918}{2 + 1 + 3} = 0.793$$

- gives an measure of the uncertainty in the branches when dividing with that attribute

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Decision Trees

- ID3: Definitions:
  - When choosing an attribute as a node:
    - Entropy comparison:
      - Split with “Body”: Entropy = 0.81
      - Split with “Tails”: Entropy = 0.79
    - The “Tails” attribute generates a partition with lower entropy (higher certainty) in the branches
    - The entropy is be calculated with the rest of the attributes
      - Choose the one with less entropy
      - This process is applied recursively on each sub-tree

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = - \sum_c (n_c/n) \log_2(n_c/n)$
      - Example:
        - $I = -(3/6)\log_2(3/6) - (3/6)\log_2(3/6)$

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n)\ \log_2(n_c/n)$
    - Examples:
      - 1 single class: $n_c = n$
        - $I = -(n_c/n)\ \log_2(n_c/n) = 0$
      - 2 classes, with the same number of samples in each: $n_c = n/2$
        - $I = -2 * (n_c/n)\ \log_2(n_c/n) = -2 * (1/2)\ \log_2(1/2) = 1$
      - 5 classes, with the same number of samples in each: $n_c = n/5$
        - $I = -5 * (n_c/n)\ \log_2(n_c/n) = -5 * (1/5) \log_2(1/5) = 2.32$
    - That is, the more distributed the samples are in different classes, the higher the entropy
      - More uncertainty, less information about the data

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = - \sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = - \sum_c (n_{ij} / n_{ij}) \log_2(n_{ijc}/n_{ij})$
    - The same operation is performed, but only among the instances with value V_j in the attribute A_i
      - Sub-table
        - Result of splitting the data with the attribute A_i
      - The entropy in that subtable is measured

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij}) \log_2(n_{ijc}/n_{ij})$
    - Example: I_{42} ?
      - Entropy of the examples that in "Body" have "Striped"

A = {Antennas, Tails, Nuclei, Body}
A_4 = Body, V_4 = {White, Striped}

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij}) \log_2(n_{ijc}/n_{ij})$
    - Example: I_{42} ?
      - Entropy of the examples that in "Body" have “Stiped”

A = {Antennas, Tails, Nuclei, Body}

A_4 = Body, V_4 = {White, Striped}

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n)\ \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij})\ \log_2(n_{ijc}/n_{ij})$
    - Example: I_{42} ?
      - Subtable with the examples that in "Body" have "Striped":

A = {Antennas, Tails, Nuclei, Body}

A_4 = Body, V_4 = {White, Striped}

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

- $I_{42} = -(3/5)\log_2\left(\frac{3}{5}\right) - (2/5)\log_2\left(\frac{2}{5}\right) = 0.971$

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij}) \log_2(n_{ijc}/n_{ij})$
    - Example: I_{41} ?
      - Entropy of the examples that in "Body" have "White"

A = {Antennas, Tails, Nuclei, Body}
A_4 = Body, V_4 = {White, Striped}

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n)\ \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij})\ \log_2(n_{ijc}/n_{ij})$
    - Example: I_{41} ?
      - Sub-table of the examples that in "Body" have "White":

A = {Antennas, Tails, Nuclei, Body}

A_4 = Body, V_4 = {White, Striped}

Example	Antennas	Tails	Nuclei	Body	Class
2	1	0	1	White	carcinogenic

- $I_{41} = 0$
  - Full certainty

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij}) \log_2(n_{ijc}/n_{ij})$
    - The same operation is performed, but only between the samples with value V_j in the attribute A_i
      - Subtable
      - The entropy in that subtable is measured
    - As before, the more distributed the samples with value V_j in the attribute A_i among the different classes, the higher the entropy
      - More uncertainty

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij}) \log_2(n_{ijc}/n_{ij})$
  - Information entropy of attribute A_i
    - $I(A_i) = \sum_j (n_{ij}/n) I_{ij}$
    - The weighted average of the entropies for each of the possible values in the attribute A_i
      - The weighted average of the entropies of the sub-tables for each value of A_i

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n)\ \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij})\ \log_2(n_{ijc}/n_{ij})$
  - Information entropy of attribute A_i
    - $I(A_i) = \sum_j (n_{ij}/n) I_{ij}$
    - Example: $I(A_4)$
      - Entropy of the "Body" attribute
      - $I(A4) = (1/6) * 0 + (5/6) * 1.971 = 0.8091$
        - Weighted average of the entropy of that attribute with each of the possible values

A = {Antennas, Tails, Nuclei, Body}
A_4 = Body, V_4 = {White, Striped}

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n) \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij}) \log_2(n_{ijc}/n_{ij})$
  - Information entropy of attribute A_i
    - $I(A_i) = \sum_j (n_{ij}/n) I_{ij}$
    - Gives a measure of the uncertainty associated with the attribute
      - The uncertainty that would remain when splitting the dataset with all values of that attribute
        - When creating the sub-tables, one for each value of the attribute

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Decision Trees

- ID3: Definitions:
  - Information entropy of a dataset
    - $I = -\sum_c (n_c/n)\ \log_2(n_c/n)$
  - Information entropy of value j of attribute A_i
    - $I_{ij} = -\sum_c (n_{ijc}/n_{ij})\ \log_2(n_{ijc}/n_{ij})$
  - Information entropy of attribute A_i
    - $I(A_i) = \sum_j (n_{ij}/n) I_{ij}$
  - Information gain of attribute A_i
    - $G(A_i) = I - I(A_i)$
      - Decrease of entropy in a dataset after splitting it using this attribute

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Decision Trees

- ID3: Example:
  - We want to generate a decision tree that classifies between normal cells and carcinogenic cells according to the data in the following table:

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of attribute A_1 (Antennas):
      - Split the table based on the values of that attribute

Example	Antennas	Tails	Nuclei	Body	Class
4	0	2	1	Striped	Normal

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
5	1	1	1	Striped	carcinogenic

Example	Antennas	Tails	Nuclei	Body	Class
6	2	2	1	Striped	carcinogenic

      - Calculate the entropies of those subtables
      - Sum of those entropies weighted by the number of instances

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_1 (Antennas)

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	**0**	**2**	**1**	**Striped**	**Normal**
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_i (Antennas)
      - $I_{10} = -\ (1/1)\log_2(1/1)\ -(0/1)\log_2(0/1) =\ 0$

Example	Antennas	Tails	Nuclei	Body	Class
4	0	2	1	Striped	Normal

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_i (Antennas)
      - $I_{10} = -(1/1)\log_2(1/1) - (0/1)\log_2(0/1) = 0$

Number of Antennas value 0, normal

Number of Antennas value 0

Example	Antennas	Tails	Nuclei	Body	Class
4	0	2	1	Striped	Normal

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_i (Antennas)
      - $I_{10} = -(1/1)\log_2(1/1) - (0/1)\log_2(0/1) = 0$

Number of Antennas value 0, normal

Number of Antennas value 0

Number of Antennas value 0, carcinogenic

Number of Antennas value 0

Example	Antennas	Tails	Nuclei	Body	Class
4	0	2	1	Striped	Normal

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_i (Antennas)
      - $I_{10} = -\ (1/1)\log_2(1/1)\ - (0/1)\log_2(0/1) =\ 0$

      - Entropy = 0:
        - Total certainty about the class in which the samples are classified

Example	Antennas	Tails	Nuclei	Body	Class
4	0	2	1	Striped	Normal

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 1 of the attribute A_1 (Antennas)

Example	Antennas	Tails	Nuclei	Body	Class
1	**1**	**0**	**2**	**Striped**	**Normal**
2	**1**	**0**	**1**	**White**	**carcinogenic**
3	**1**	**2**	**0**	**Striped**	**Normal**
4	0	2	1	Striped	Normal
5	**1**	**1**	**1**	**Striped**	**carcinogenic**
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 1 of the attribute A_1 (Antennas)
      - $I_{11} = -(2/4)\log_2(2/4) - (2/4)\log_2(2/4) = 1$

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
5	1	1	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 1 of the attribute A_1 (Antennas)
      - $I_{11} = -(2/4)\log_2(2/4) - (2/4)\log_2(2/4) = 1$

Number of Antennas value 1, normal

Number of Antennas value 1

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
5	1	1	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 1 of the attribute A_1 (Antennas)
      - $I_{11} = -(2/4)\log_2(2/4) - (2/4)\log_2(2/4) = 1$

Number of Antennas value 1, normal

Number of Antennas value 1

Number of Antennas value 1, carcinogenic

Number of Antennas value 1

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
5	1	1	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 1 of the attribute A_1 (Antennas)
      - $I_{11} = -(2/4)\log_2(2/4) - (2/4)\log_2(2/4) = 1$
      - Entropy = 1:
        - Highest entropy (uncertainty) that can be had with 2 classes
          - The samples are distributed into the two classes with 50% each
          - If there were more samples of one class than another, there would be more certainty, less entropy

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
5	1	1	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 2 of the attribute A_1 (Antennas)

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	**2**	**2**	**1**	**Striped**	**carcinogenic**

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 2 of the attribute A_1 (Antennas)
      - $I_{12} = -(0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$

Number of Antennas value 2, normal

Number of Antennas value 2

Number of Antennas value 2, carcinogenic

Number of Antennas value 2

Example	Antennas	Tails	Nuclei	Body	Class
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_1 (Antennas)
      - $\mathrm{I}_{10} = -(1/1)\log_2(1/1) - (0/1)\log_2(0/1) = 0$
    - Information entropy of the value 1 of the attribute A_1 (Antennas)
      - $\mathrm{I}_{11} = -(2/4)\log_2(2/4) - (2/4)\log_2(2/4) = 1$
    - Information entropy of the value 2 of the attribute A_1 (Antennas)
      - $\mathrm{I}_{12} = -(0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of attribute A_1 (Antennas)
      - $I(A_1) = \Sigma\,(n_{ij}/n)I_{ij} = (1/6)\cdot I_{10} + (4/6)\cdot I_{11} + (1/6)\cdot I_{12} = 0.66$

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of attribute A_1 (Antennas)
      - $I(A_1) = \Sigma\,(n_{ij}/n)I_{ij} = (1/6)\cdot I_{10} + (4/6)\cdot I_{11} + (1/6)\cdot I_{12} = 0.66$

Number of Antennas value 0 Number of Antennas value 1 Number of Antennas value 2

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of attribute A_1 (Antennas)
      - $I(A_1) = \Sigma\,(n_{ij}/n) I_{ij} = (1/6) \cdot I_{10} + (4/6) \cdot I_{11} + (1/6) \cdot I_{12} = 0.66$

Number of Antennas value 0

Number of Antennas value 1

Number of Antennas value 2

Number of Antennas (instances)

Example	Antennas	Tails	Nuclei	Body	Class
1	1	0	2	Striped	Normal
2	1	0	1	White	carcinogenic
3	1	2	0	Striped	Normal
4	0	2	1	Striped	Normal
5	1	1	1	Striped	carcinogenic
6	2	2	1	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_2 (Tails)
      - $\mathrm{I}_{20} = -(1/2)\log_2(1/2) - (1/2)\log_2(1/2) = 1$
    - Information entropy of the value 1 of the attribute A_2 (Tails)
      - $\mathrm{I}_{21} = -(0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$
    - Information entropy of the value 2 of the attribute A_2 (Tails)
      - $\mathrm{I}_{22} = -(2/3)\log_2(2/3) - (1/3)\log_2(1/3) = 0.9183$
    - Information entropy of attribute A_2 (Tails)
      - $I(A_1) = \Sigma\,(n_{\mathrm{ij}}/n)I_{\mathrm{ij}} = (2/6)\cdot I_{20} + (1/6)\cdot I_{21} + (3/6)\cdot I_{22} = 0.79$

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_3 (Nuclei)
      - $\mathrm{I}_{30} = -(1/1)\log_2(1/1) - (0/1)\log_2(0/1) = 0$
    - Information entropy of the value 1 of the attribute A_3 (Nuclei)
      - $\mathrm{I}_{31} = -(1/4)\log_2(1/4) - (3/4)\log_2(3/4) = 0.8113$
    - Information entropy of the value 2 of the attribute A_3 (Nuclei)
      - $\mathrm{I}_{32} = -(1/1)\log_2(1/1) - (0/1)\log_2(0/1) = 0$
    - Information entropy of attribute A_3 (Nuclei)
      - $I(A_1) = \Sigma\,(n_{\mathrm{ij}}/n)I_{\mathrm{ij}} = (1/6)\cdot I_{30} + (4/6)\cdot I_{31} + (1/6)\cdot I_{32} = 0.54$

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value "White" of the attribute A_4 (Body)
      - $\mathrm{I}_{40} = -\ (0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$
    - Information entropy of the value "Stripped" of the attribute A_4 (Body)
      - $\mathrm{I}_{41} = -\ (2/5)\log_2(2/5) - (3/5)\log_2(3/5) = 0.9710$
    - Information entropy of attribute A_4 (Body)
      - $I(A_1) = \Sigma\,(n_{\mathrm{ij}}/n)I_{\mathrm{ij}} = (1/6)\cdot I_{40} + (5/6)\cdot I_{41} = 0.81$

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Decision Trees

- ID3: Example:
  - The attribute with the lowest entropy is chosen:
    - The entropy of each attribute determines the level of uncertainty when separating the dataset with all the values of that attribute
    - The attribute with lower entropy indicates that by separating the samples with that attribute, each of the resulting sets will have more certainty in the data.
      - That is, the instances of the different classes will be further unified in each of those sets (each of the values of that attribute)
      - There will be less disparity of instances of different classes in each set
      - That is, it better explains the sets
    - Attribute with lowest Entropy: Nuclei

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Decision Trees

- ID3: Example:
  - The attribute with the lowest entropy is chosen: Nuclei
  - The dataset is split and that attribute is removed:
    - Nuclei = 0

Example	Antennas	Tails	Body	Class
3	1	2	Striped	Normal

    - Nuclei = 1

Example	Antennas	Tails	Body	Class
2	1	0	White	carcinogenic
4	0	2	Striped	Normal
5	1	1	Striped	carcinogenic
6	2	2	Striped	carcinogenic

    - Nuclei = 2

Example	Antennas	Tails	Body	Class
1	1	0	Striped	Normal

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Decision Trees

- ID3: Example:

Nuclei

0

Example	Antennas	Tails	Body	Class
3	1	2	Striped	Normal

2

Example	Antennas	Tails	Body	Class
1	1	0	Striped	Normal

1

Example	Antennas	Tails	Body	Class
2	1	0	White	carcinogenic
4	0	2	Striped	Normal
5	1	1	Striped	carcinogenic
6	2	2	Striped	carcinogenic

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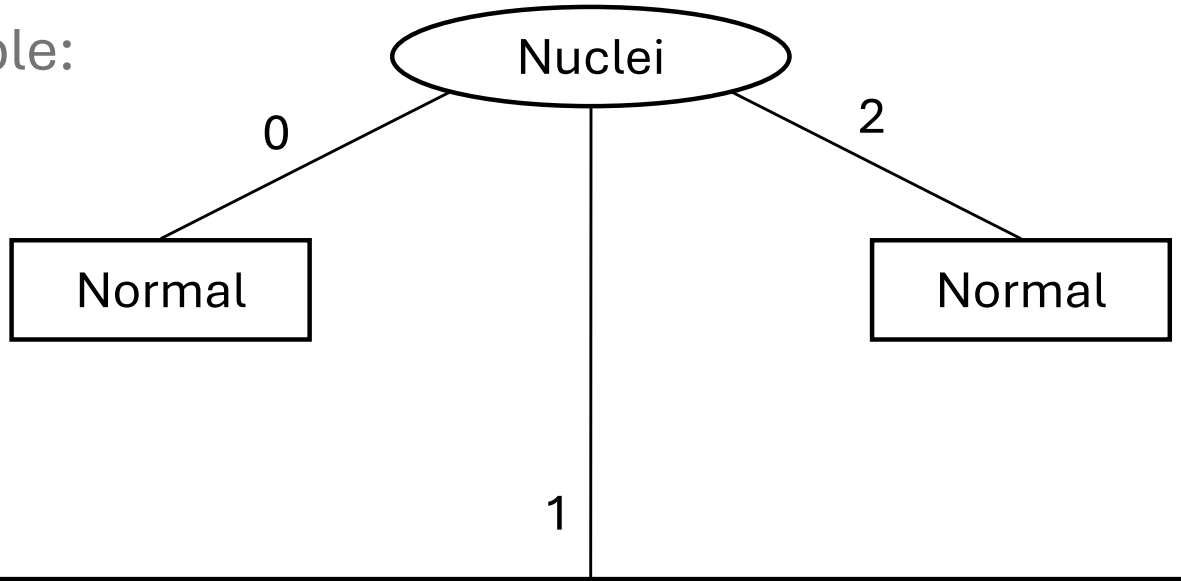
Decision Trees

- ID3: Example:



Example	Antennas	Tails	Body	Class
2	1	0	White	carcinogenic
4	0	2	Striped	Normal
5	1	1	Striped	carcinogenic
6	2	2	Striped	carcinogenic

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_1 (Antennas)
      - $\mathrm{I}_{10} = -(1/1)\log_2(1/1) - (0/1)\log_2(0/1) = 0$
    - Information entropy of the value 1 of the attribute A_1 (Antennas)
      - $\mathrm{I}_{11} = -(0/2)\log_2(0/2) - (2/2)\log_2(2/2) = 0$
    - Information entropy of the value 2 of the attribute A_1 (Antennas)
      - $\mathrm{I}_{22} = -(0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$
    - Information entropy of attribute A_2 (Tails)
      - $I(A_1) = \Sigma\,(n_{\mathrm{ij}}/n)I_{\mathrm{ij}} = (1/4)\cdot I_{10} + (2/4)\cdot I_{11} + (1/4)\cdot I_{12} = 0$

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value 0 of the attribute A_2 (Tails)
      - $\mathrm{I}_{20} = -\ (0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$
    - Information entropy of the value 1 of the attribute A_2 (Tails)
      - $\mathrm{I}_{21} = -\ (0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$
    - Information entropy of the value 2 of the attribute A_2 (Tails)
      - $\mathrm{I}_{22} = -\ (1/2)\log_2(1/2) - (1/2)\log_2(1/2) = 1$
    - Information entropy of attribute A_2 (Tails)
      - $I(A_1) = \Sigma\,(n_{\mathrm{ij}}/n)I_{\mathrm{ij}} = (1/4)\cdot I_{20} + (1/4)\cdot I_{21} + (2/4)\cdot I_{22} = 0.5$

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Decision Trees

- ID3: Example:
  - Entropies are calculated:
    - Information entropy of the value "White" of the attribute A_4 (Body)
      - $\mathrm{I}_{40} = -\ (0/1)\log_2(0/1) - (1/1)\log_2(1/1) = 0$
    - Information entropy of the value "Striped" of the attribute A_4 (Body)
      - $\mathrm{I}_{41} = -\ (1/3)\log_2(1/3) - (2/3)\log_2(2/3) = 0.9183$
    - Information entropy of attribute A_3 (Nuclei)
      - $I(A_1) = \Sigma\,(n_{\mathrm{ij}}/n)I_{\mathrm{ij}} = (1/4)\cdot I_{40} + (3/4)\cdot I_{41} = 0.6887$

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Decision Trees

- ID3: Example:
  - The attribute with the lowest entropy is chosen: Antennas
  - The dataset is split and that attribute is removed:
    - Antennas = 0

Example	Tails	Body	Class
4	2	Striped	Normal

    - Antennas = 1

Example	Tails	Body	Class
2	0	White	carcinogenic
5	1	Striped	carcinogenic

    - Antennas = 2

Example	Tails	Body	Class
6	2	Striped	carcinogenic

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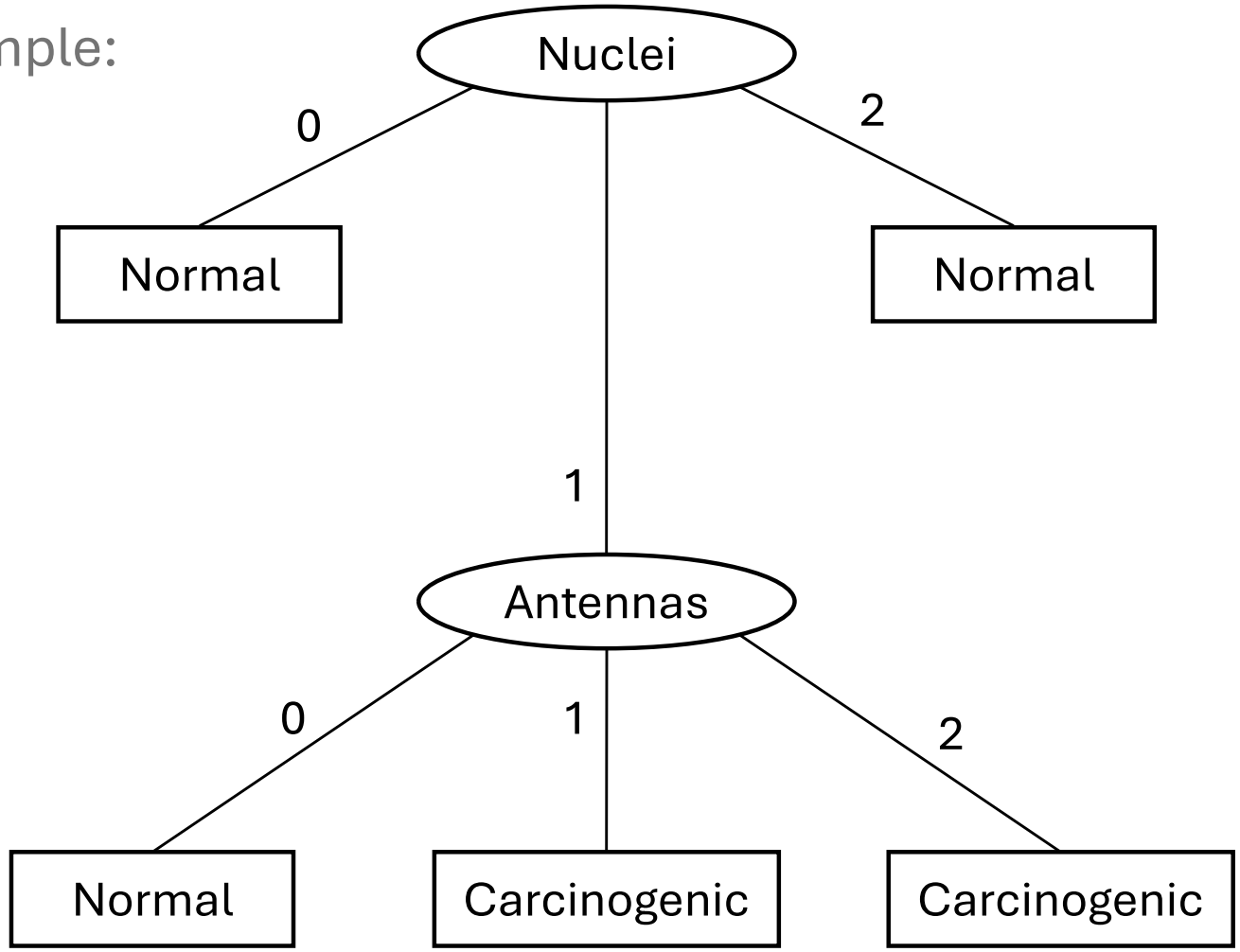
Decision Trees

- ID3: Example:



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Decision Trees

- ID3: Some practical questions:
  - How much to grow the tree?
  - Is this attribute selection measure appropiate?
  - How to handle training data with missing (unknown) values in attributes?
  - How to handle attributes with different costs?
  - How to handle continuous attributes?

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Decision Trees

- ID3: How much to grow the tree?
  - If the tree is grown until it correctly classifies all training examples: overfitting
    - If there is noise in the examples, the noise is learned
    - If there are few examples associated with the leaf nodes, they are not representative of the true function
  - ID3 produces trees that overfit training examples, and do not work properly with new examples
    - The model is not able to generalize

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Decision Trees

- ID3: Solutions to overfitting :
  - Pre-pruning:
  - Post-pruning:

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Decision Trees

- ID3: Solutions to overfitting :
  - Pre-pruning:
    - Stop growing the tree before it reaches the point where it perfectly classifies the training examples
    - When to do it?

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Decision Trees

- ID3: Solutions to overfitting :
  - Pre-pruning:
    - Stop growing the tree before it reaches the point where it perfectly classifies the training examples
    - When to do it?
      - Apply a statistical test to estimate whether expanding a particular node is likely to produce an improvement beyond the training set

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Decision Trees

- ID3: Solutions to overfitting :
  - Pre-pruning:
    - Stop growing the tree before it reaches the point where it perfectly classifies the training examples
    - When to do it?
  - Post-pruning:
    - Allow it to overfit the data, and then prune it by replacing subtrees with a leaf
    - Requires higher computational cost, but offers better results

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Decision Trees

- ID3: Solutions to overfitting : Post-pruning:
  - Starting from the bottom, examine the subtrees of the non-terminal nodes
  - Pruning a node entails:
    - 1.- Delete its corresponding subtree with root in that node
    - 2.- Convert the node to a leaf node
    - 3.- Assign the most common classification of the training examples associated to that node
  - Prune only if the resulting pruned tree improves or equals the performance of the original tree
    - Test (validation)
  - The tree is pruned iteratively, always choosing the node whose pruning implies the greatest improvement, until there is no improvement

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Decision Trees

- ID3: Is this attribute selection measure adequate?
  - The ID3 algorithm favors choosing attributes with many values
    - For example, attribute "Date" in a set with the examples on different days
      - This attribute perfectly characterizes each training example, and we would have a perfect tree of depth 1, with one leaf node per sample
      - The gain information is the highest
        - Having so many branches forces the examples to be separated into very small sets
      - But it would be very bad with unseen examples (test)

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Decision Trees

- ID3: Is this attribute selection measure appropriate?
  - There are other measures instead of information gain
    - Gain ratio: penalizes attributes with many values and uniformly distributed

$$\frac{G(A_i)}{H(A_i)}$$

    - where $H(A_i)$ denotes the entropy with respect to the values of A_i

$$H(A_i) = -\sum_j \frac{n_{ij}}{n} \log_2 \frac{n_{ij}}{n}$$

      - It does not take into account the classes, only the different values it takes
      - When an attribute has many different values, it has a very large entropy, so it is divided by a larger value.

    - The gain ratio favors those attributes that, with equal gain, separate the data into fewer subsets

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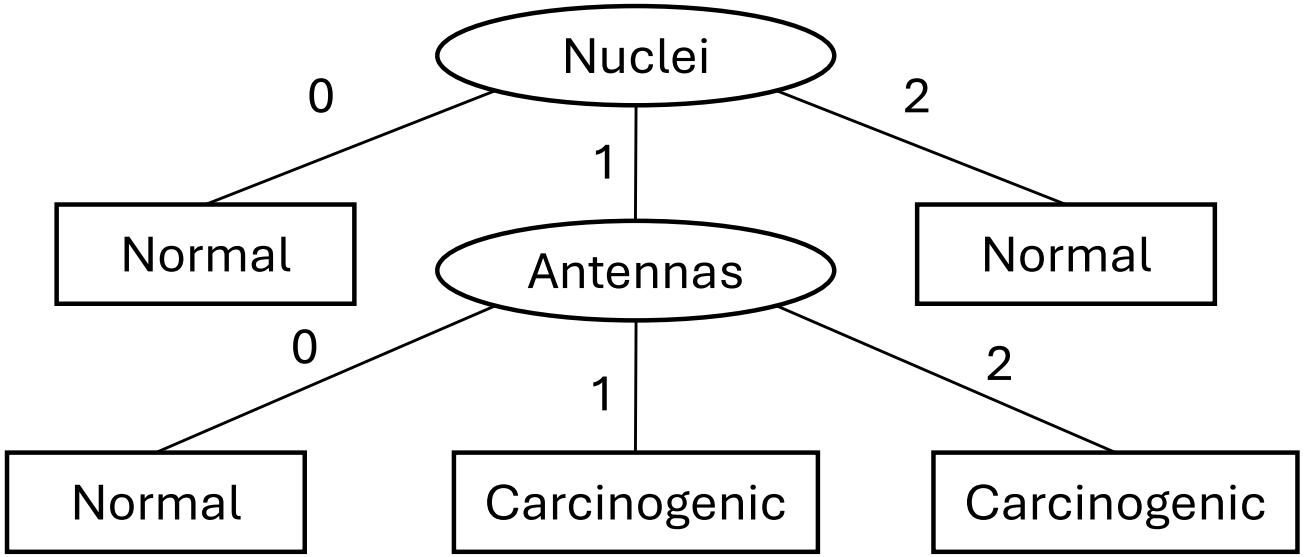
Decision Trees

- ID3: How to handle attributes with missing values?
  - When classifying a new case with a missing value:
    - Instances with unknown values are classified according to the highest probability provided by the tree
      - Example:
        - Nuclei = ?
        - Antennas = 1



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Decision Trees

- ID3: How to handle attributes with different costs?
  - Are all attributes equally valuable when classifying?
  - Enter the cost in the attribute selection measure
    - For example:

$$\frac{G(A_j)}{Cost\ (A_j)} \qquad \frac{2^{G(A_j)}-1}{(Cost\ (A_j)+1)^w}$$

w: constant between 0 and 1 that weights the importance of cost versus profit

  - They try to place the low-cost ones at the top of the tree, only calling the high-cost ones when necessary
    - That is, when we want to improve the classification

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Decision Trees

- ID3: How to handle continuous attributes?
  - They are discretized:
    - They are partitioned into a discrete set of intervals
      - Example: 2 intervals: $A < c$ and $A \geq c$
    - Find a value of c with the highest information gain
      - First, order the examples by the attribute A
        - This value is always where the value of the class changes

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Decision Trees

- ID3: How to handle continuous attributes?
  - They are discretized:
    - Process:
      - Order the examples according to the continuous value of A
      - Take each pair of contiguous values of A with a different value of C and find the average
      - Among these candidates to be c, choose the one that returns the highest information at this time
      - Once c is chosen, this new attribute competes with the other discrete ones to see if it is the chosen one
      - If it is not chosen, a different c may come out later
        - That is, the attribute is created dynamically every time the attribute is chosen.

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Decision Trees

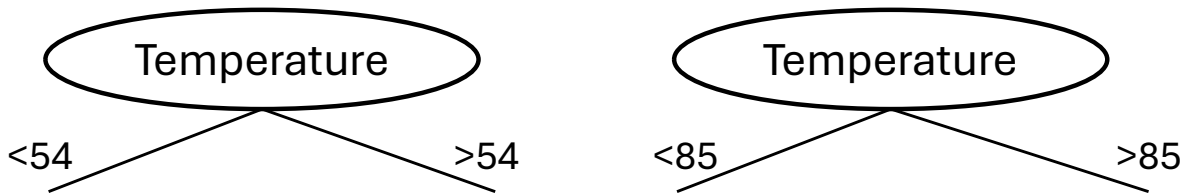
- ID3: How to handle continuous attributes?
  - Example:

Temperature	80	48	90	72	40	60
Class	A	B	B	A	B	A

Ordered:

Temperature	40	48	60	72	80	90
Class	B	B	A	A	A	B

- Candidates are $c_1 = (48 + 60)/2 = 54$ and $c_2 = (80 + 90)/2 = 85$
- It would be necessary to calculate the gain for



  - And these two gains have to compete against the rest of the attributes

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Decision Trees

- ID3: How to handle continuous attributes?
  - Example:

Temperature	80	48	90	72	40	60
Class	A	B	B	A	B	A

Ordered:

Temperature	40	48	60	72	80	90
Class	B	B	A	A	A	B

- Candidates are $c_1 = (48 + 60)/2 = 54$ and $c_2 = (80 + 90)/2 = 85$
- Other alternatives:
  - More than two intervals
  - Use linear combinations of various attributes: $\alpha A + \beta B > c$ ...

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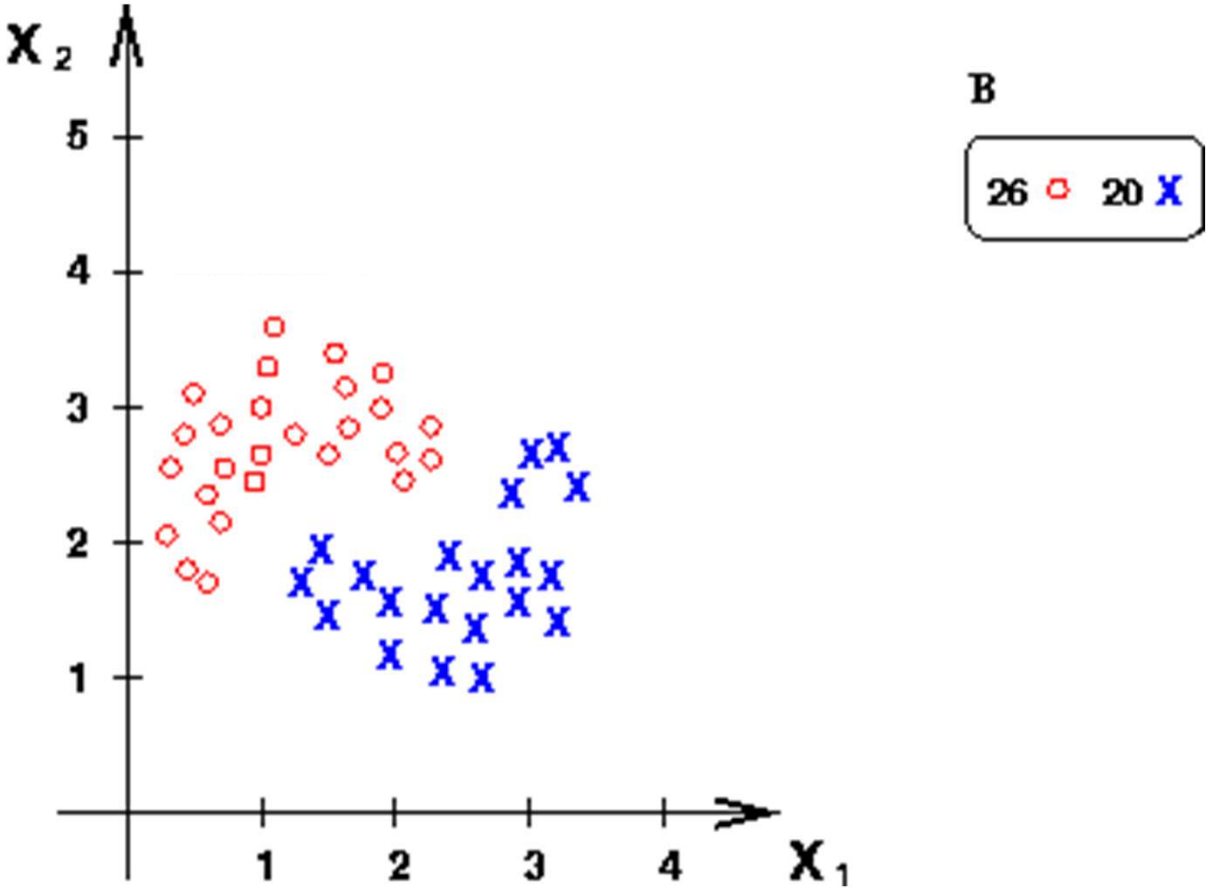
Decision Trees

- Example:
  - Continuous Attributes:



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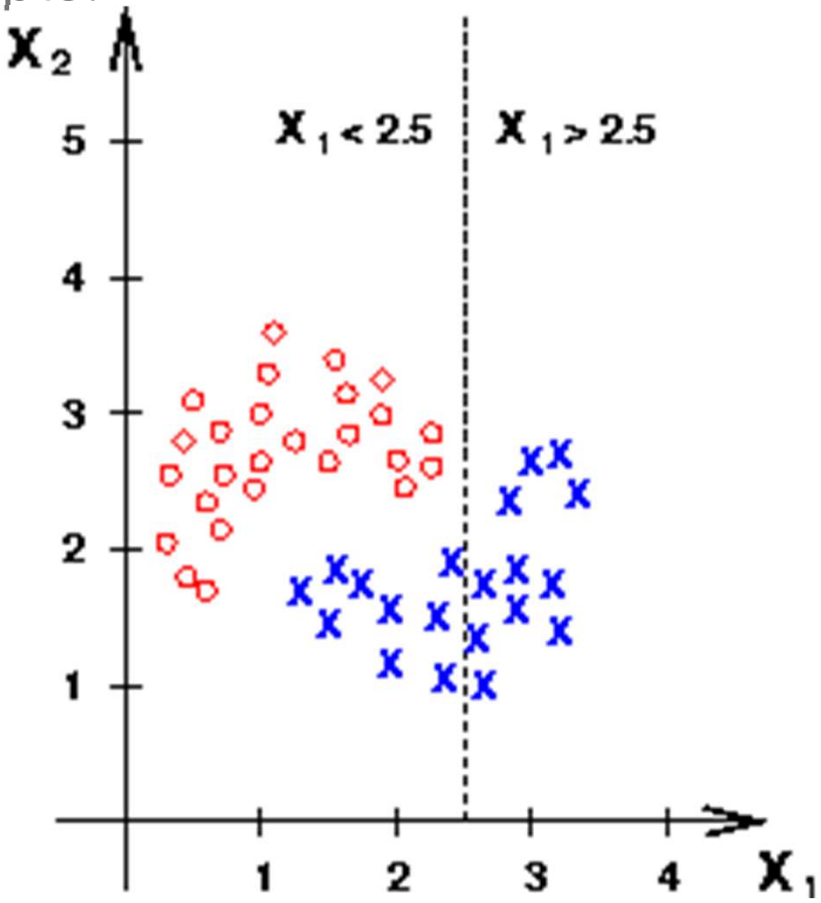
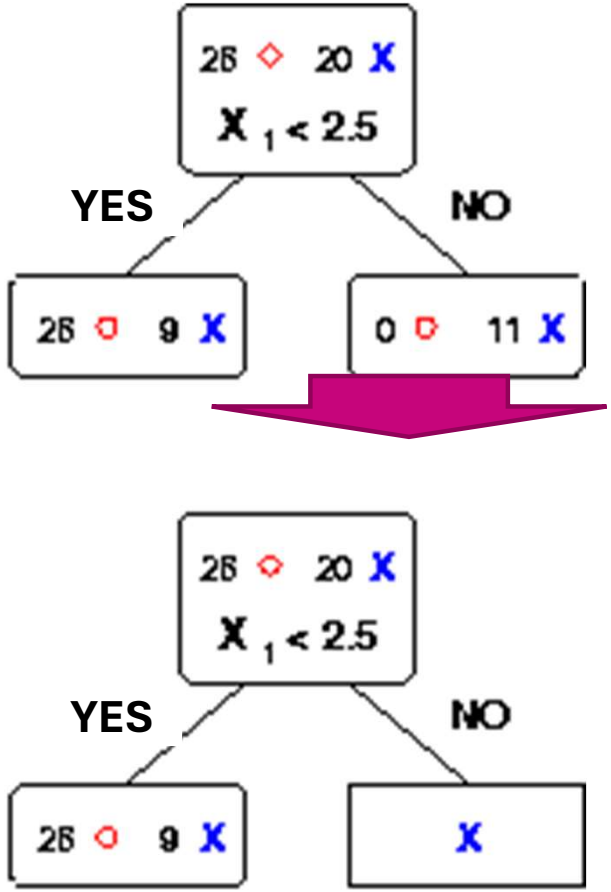
Decision Trees

- Example:





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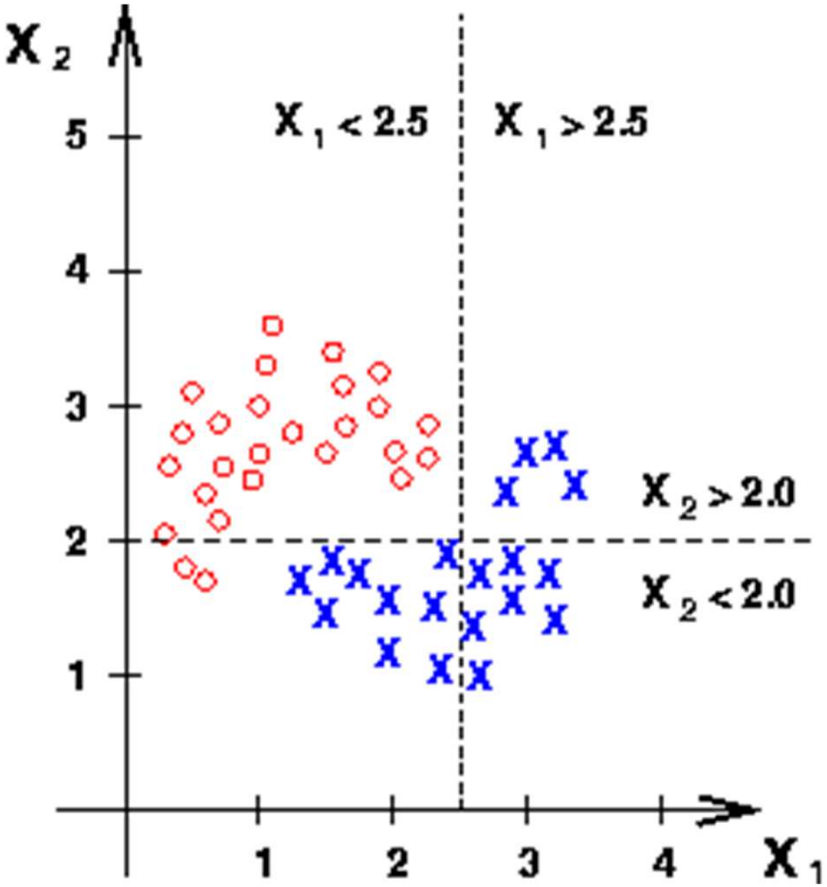
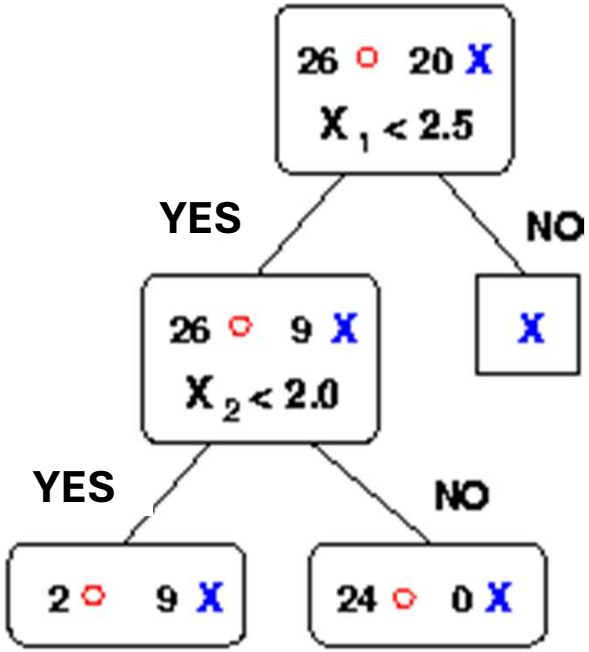
Decision Trees

- Example:





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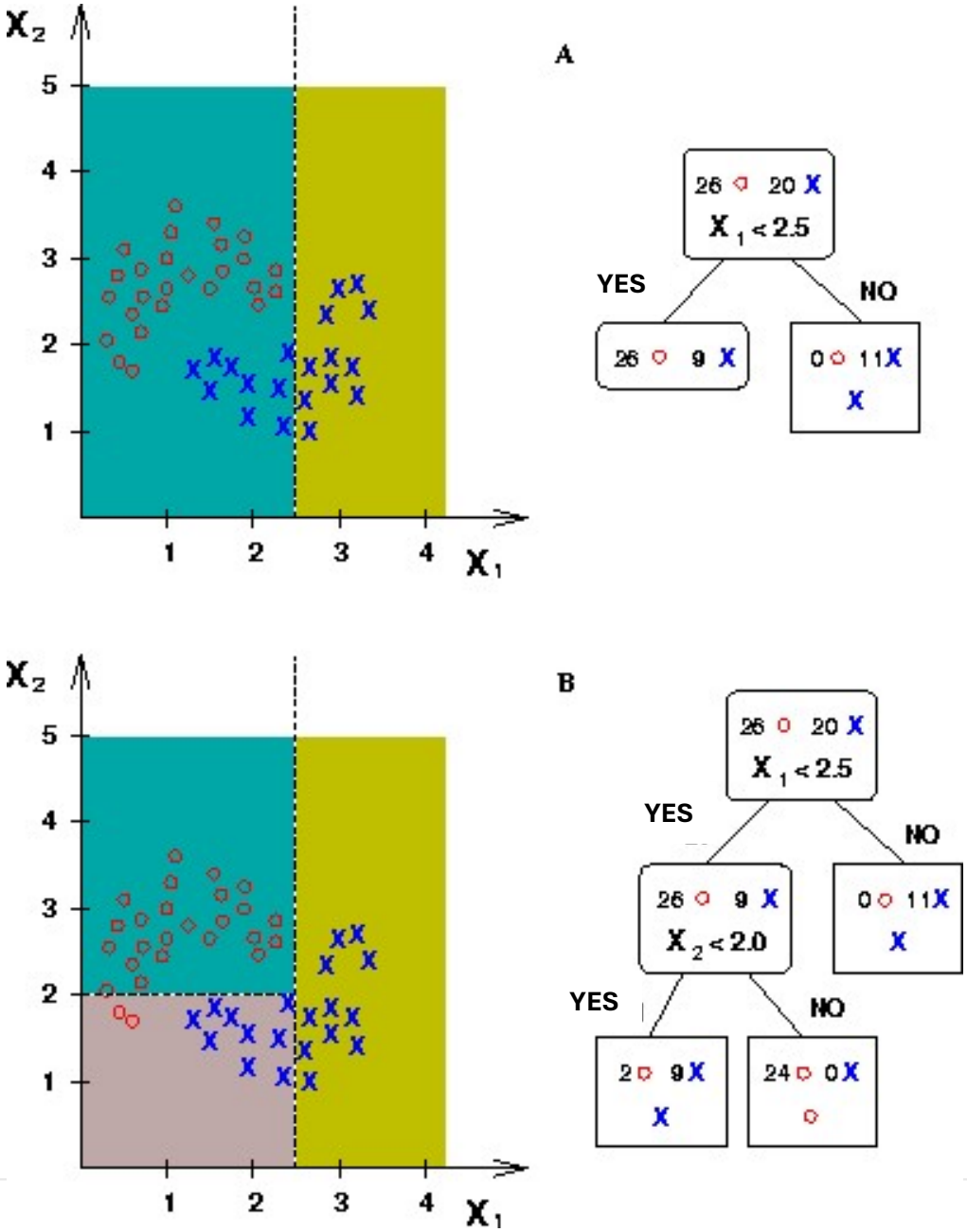
Decision Trees

- Example:
  - Regions:
    - Parallelepipeds



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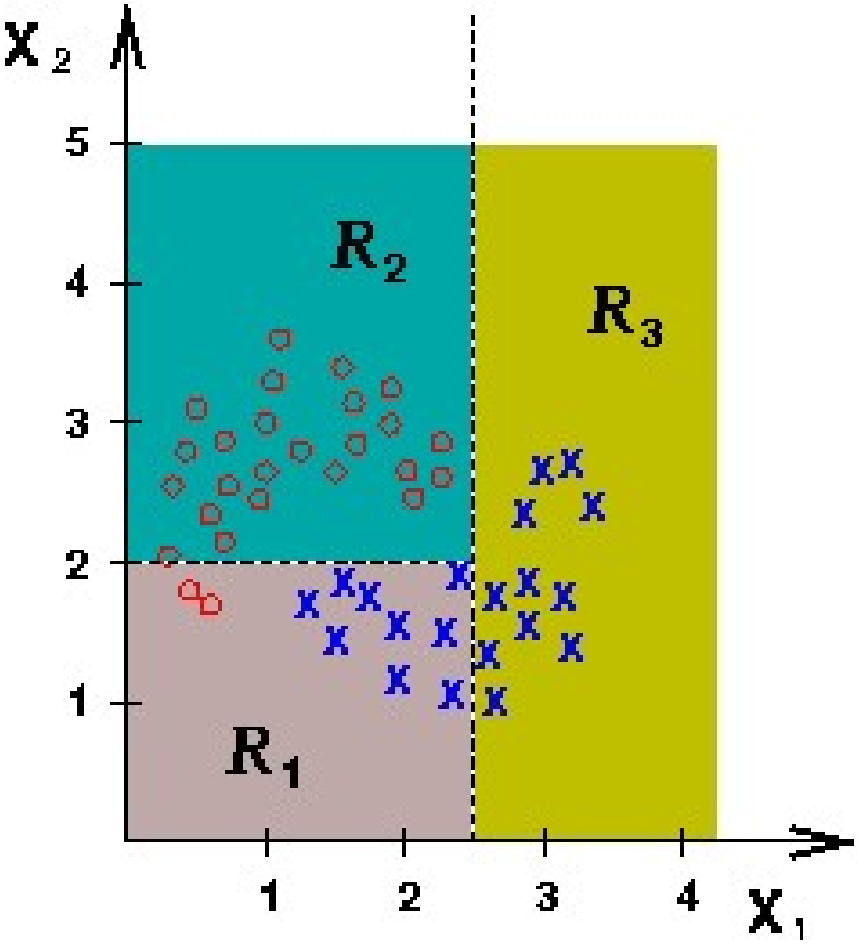
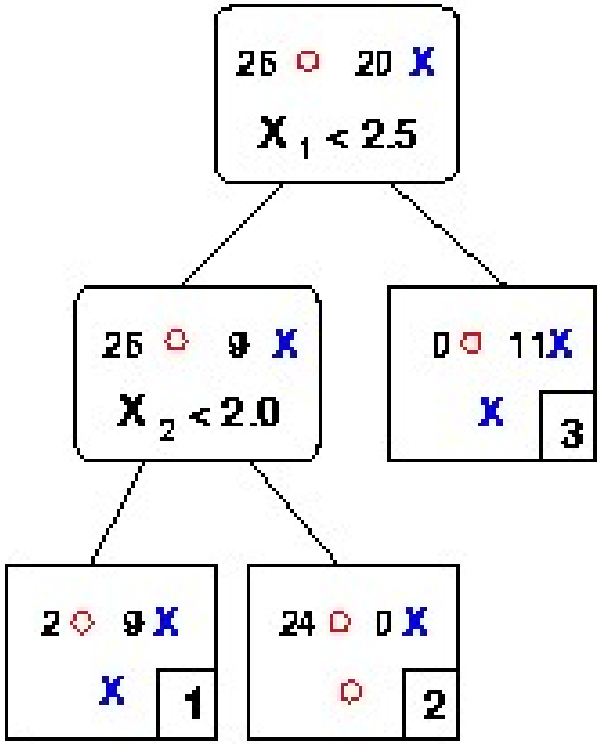
Decision Trees

- Example:
  - Each leaf node has a region associated with it:





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Decision Trees

- Example:
  - Continuous attributes
    - Allowing a condition to be a linear combination of attributes

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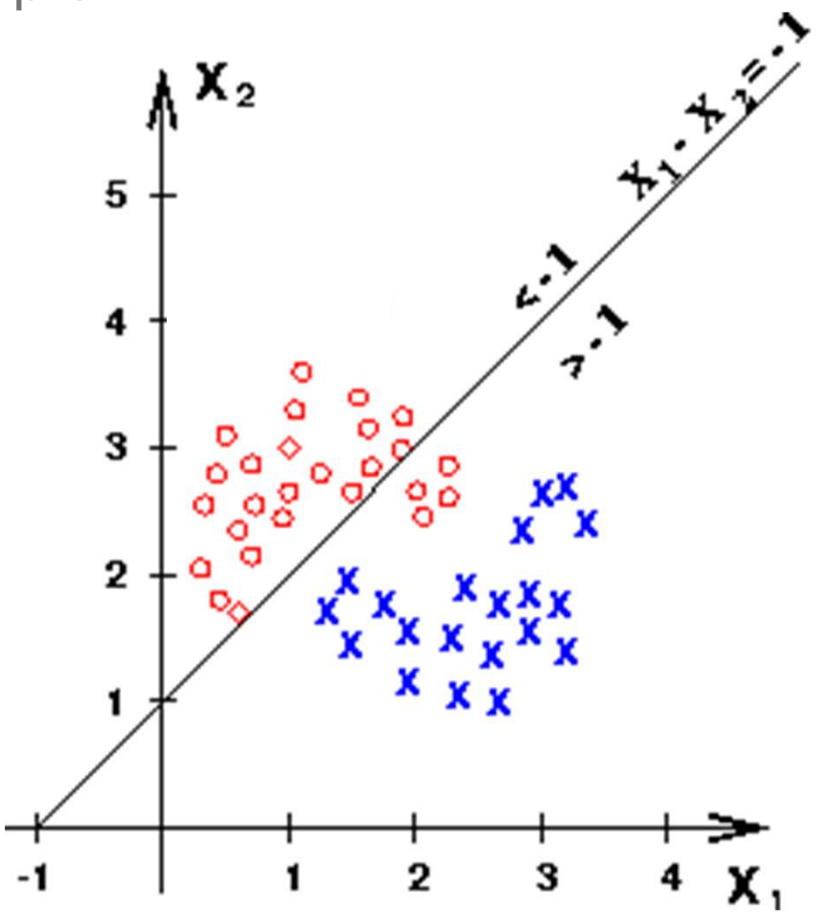
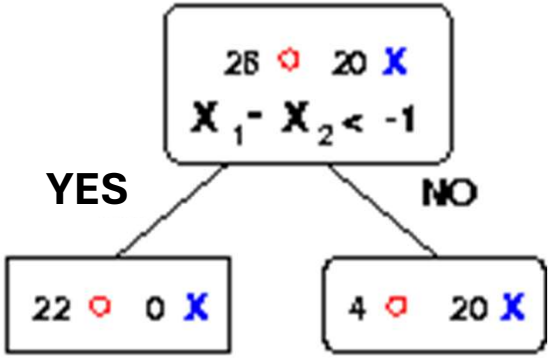
Decision Trees

- Example:





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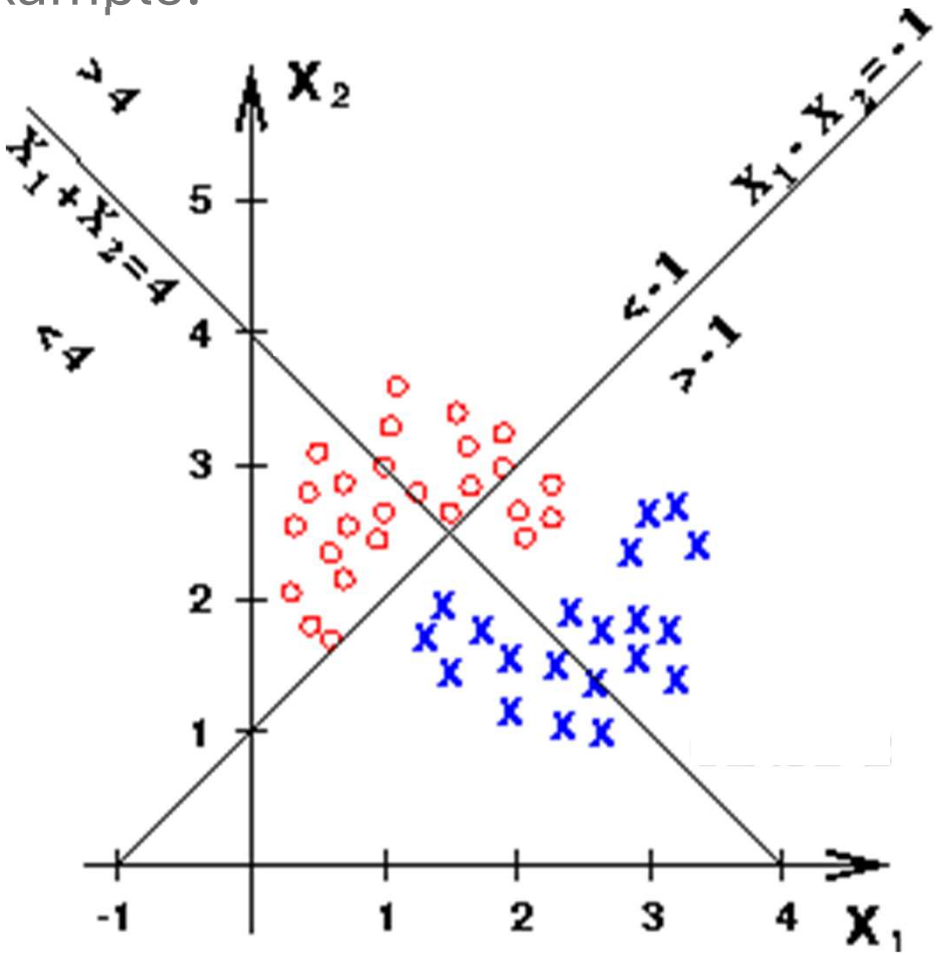
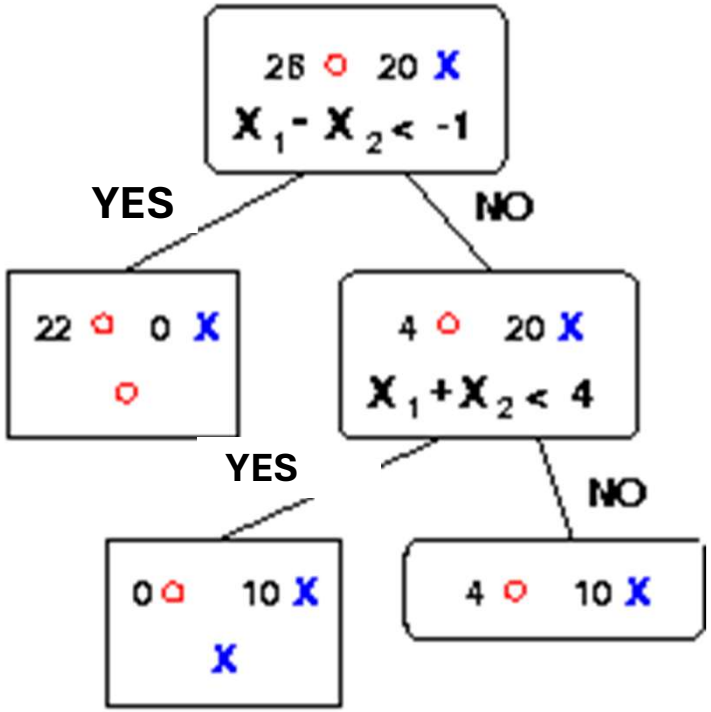
Decision Trees

- Example:





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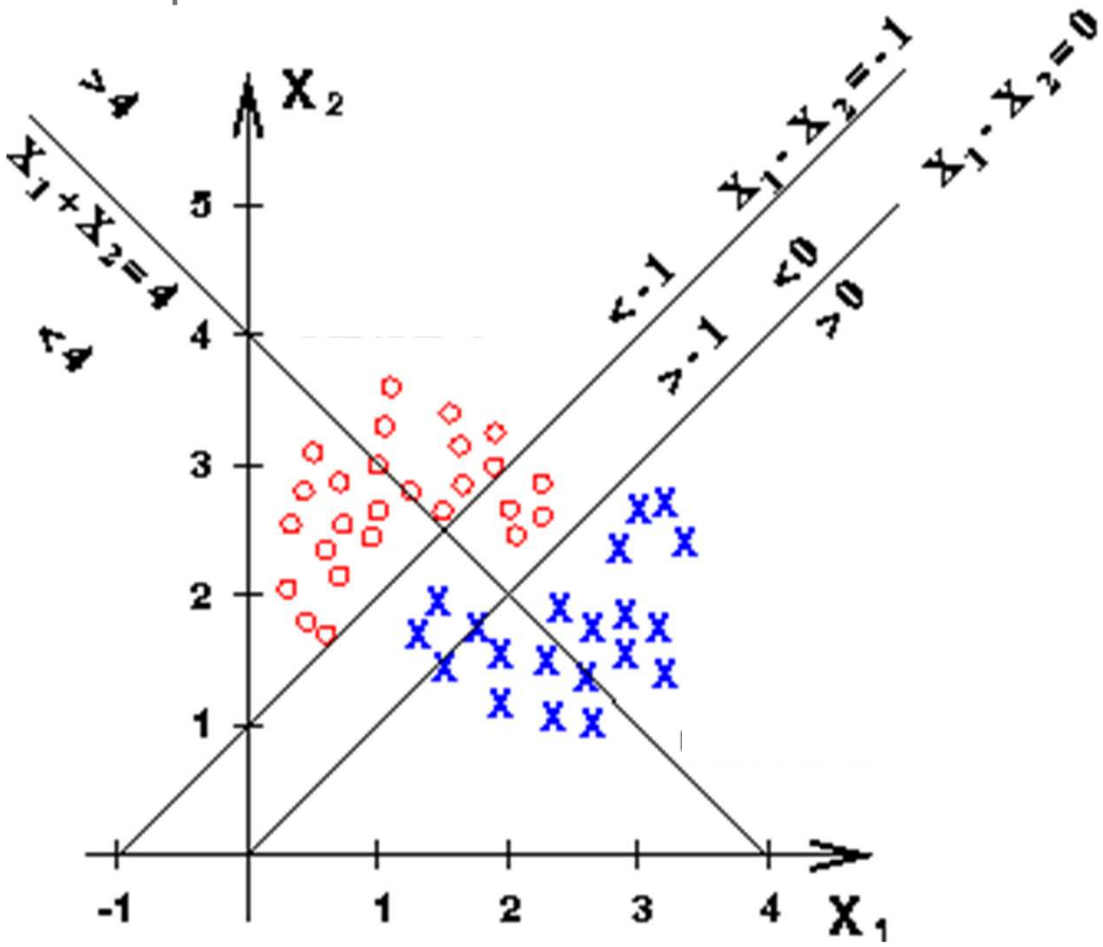
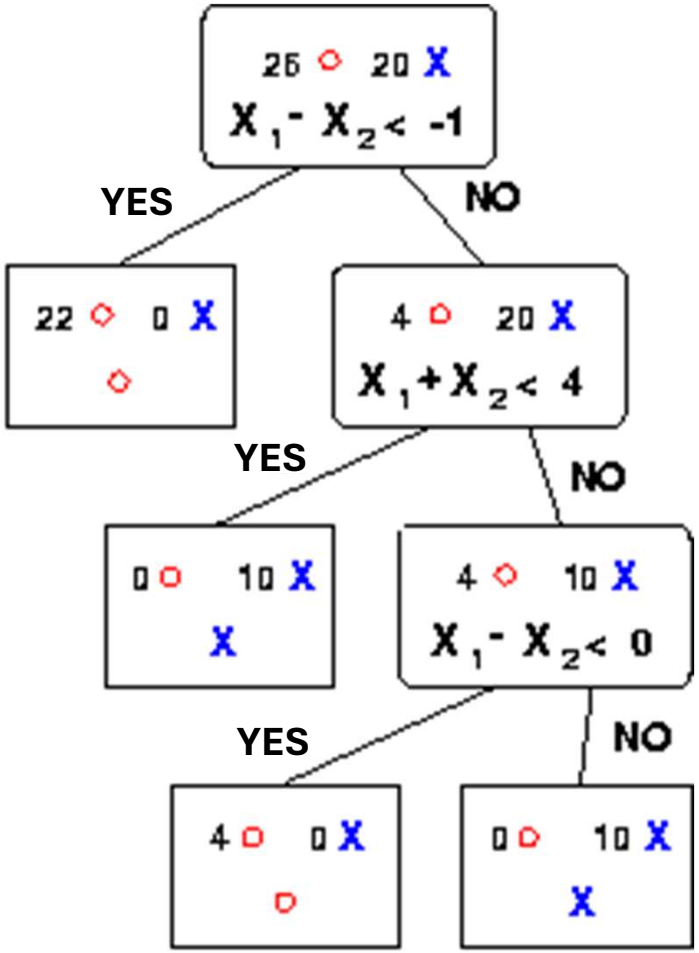
Decision Trees

- Example:





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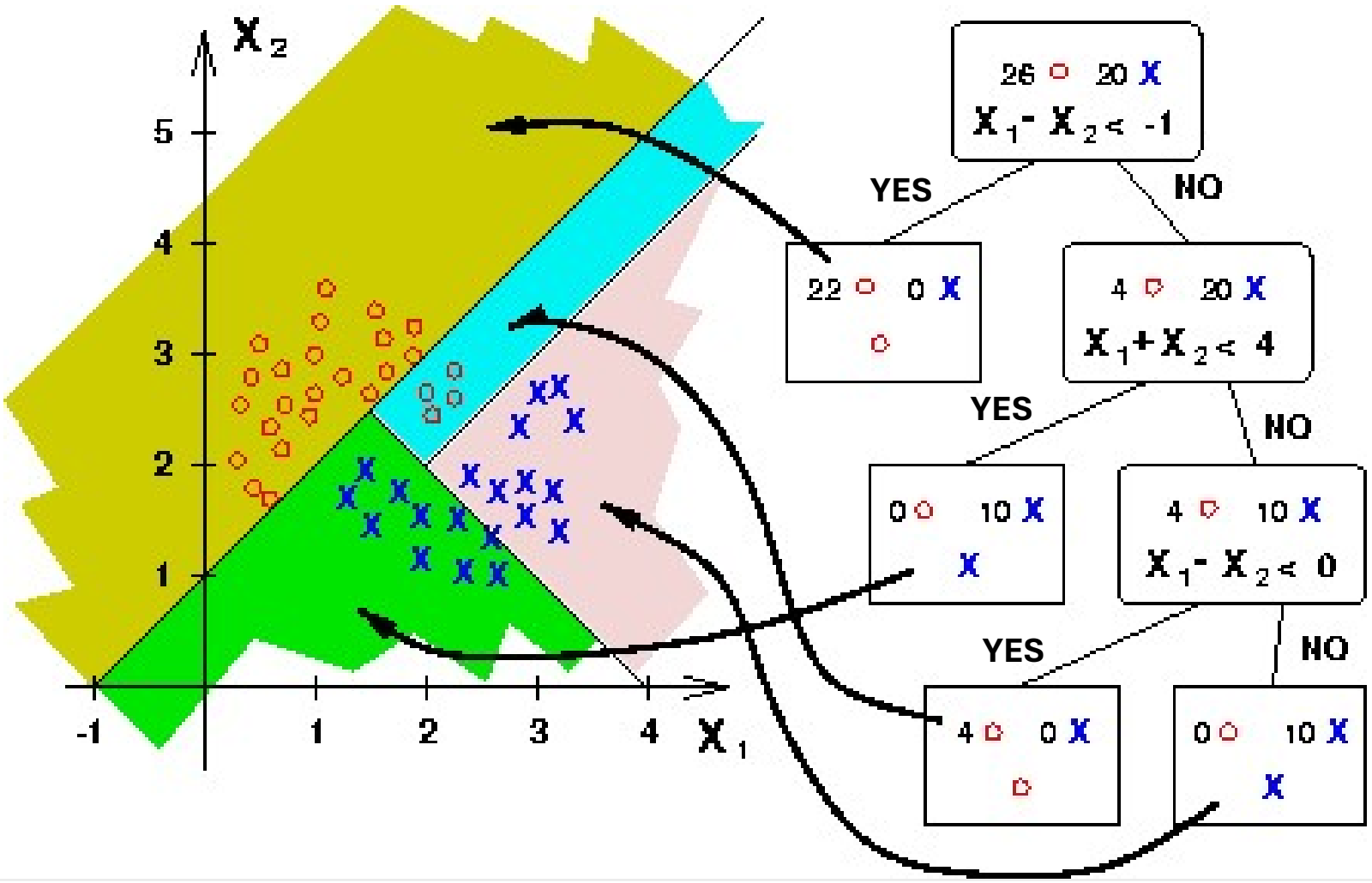
Decision Trees

- Example:
  - Each leaf node has an associated region.



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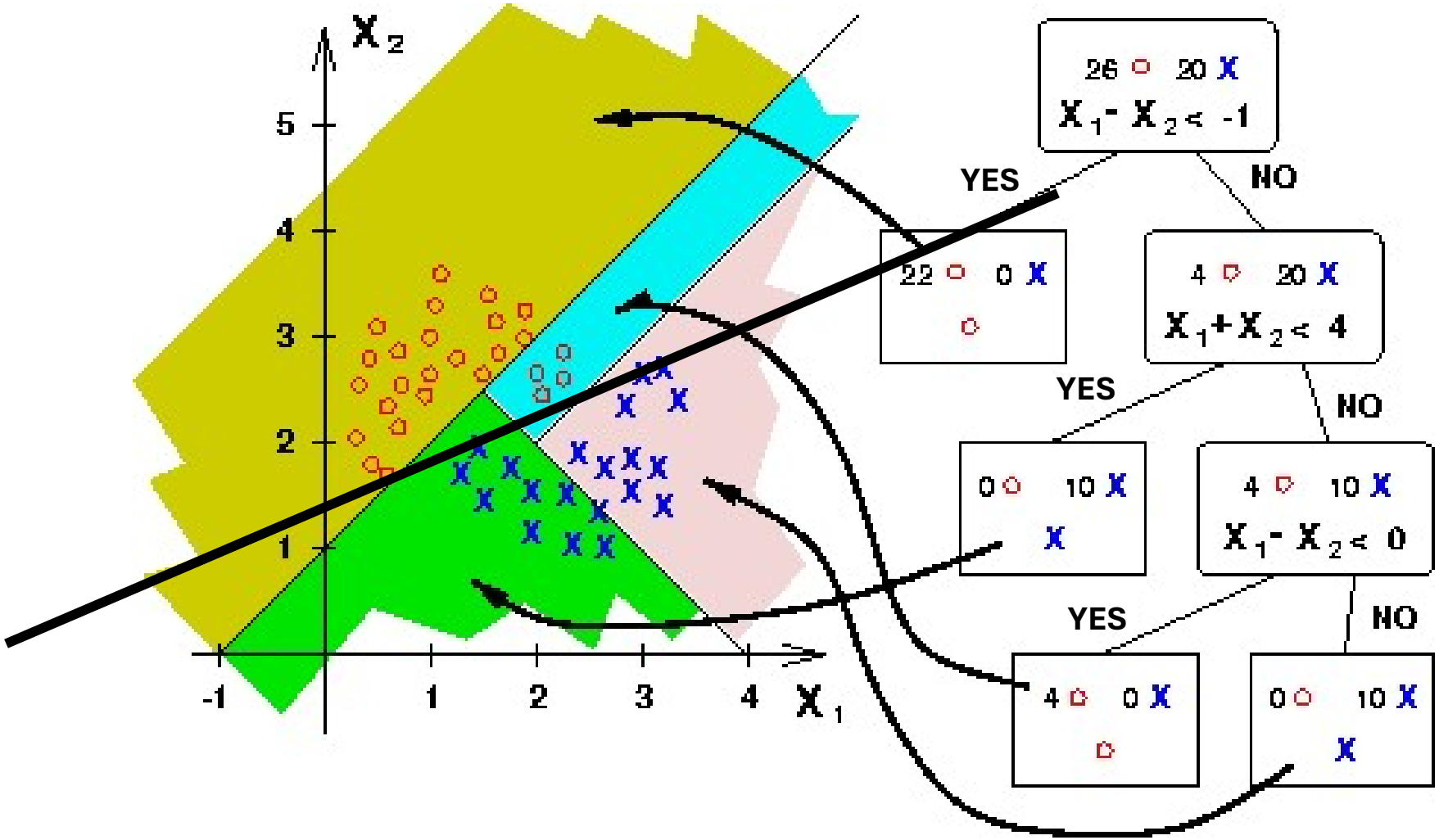
Decision Trees

- Example:
  - Overfitted tree vs. Linear classification:



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Decision Trees

- C4.5: (Quinlan, 1993)
  - ID3-based algorithm with all the above modifications:
    - Allows working with continuous values for attributes
    - Choose attributes using the gain ratio
      - Eliminate ID3's tendency to pick attributes with many values
    - Incorporates post-pruning rules:
      - Generates 1 rule per path
      - Eliminate antecedents of each rule whenever the error is better or equal
    - Enter costs in the attributes
    - Handle unknown values
    - Order the rules by their estimated error (least to most)
      - This sequence is considered when classifying instances

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Decision Trees

- CART:
  - Classification And Regression trees
  - Breiman and others, 1984
  - 4 phases:
    - Tree construction
    - Stop of the growth process
      - A maximum tree is constituted that overfits the information contained in the dataset
    - Pruning
      - Makes the tree simpler and leaves only the most important nodes
    - Selection of the optimal tree with generalization capacity

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Decision Trees

- CART:
  - How to choose an attribute to be a node:
    - Looking for a measure of its purity
      - The attribute with the highest purity is chosen.
    - A dataset is purer when the instances are less spread out among the classes
      - Lower entropy
      - Greater certainty
    - A partition function must be used to ensure that the purity at the child nodes is maximized
      - Children should be as pure as possible
        - The instances in the children should have as much class homogeneity as possible

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Decision Trees

- CART:
  - How to choose an attribute to be a node:
    - Instead of measuring the purity at the nodes, the impurity is measured
    - The measure of the impurity:
      - Should be maximum when the instances of a node are equally divided among all classes
      - Must be 0 when the instances of a node are of the same class
    - To measure this impurity, we could use:
      - Entropy (ID3, C4.5)
      - Gini index (CART)

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Decision Trees

- CART:
  - Gini index:
    - The Gini index tells us what is the probability of misclassifying an observation
      - Measurement of the impurity in a node
    - The Gini index it is a measure of how many times an element selected randomly from the dataset would be incorrectly labelled if it were randonmly labelled according to the distribution of labels in the dataset
    - That is, for a dataset, the Gini index measures
      - A sample is randomly chosen
      - Calculate the probability of making a mistake when labeling it according to the distribution of labels in the set
      - This process is done for all of the samples of each class

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Decision Trees

- CART:
  - Gini index:
    - It is calculated by adding the probability of choosing each instance multiplied for the probability of wrongly labeling it

      - For example, if we have 3 classes, with 10 instances:
        - A, A, A, A, B, B, B, C, C, C

      - For class A:
        - Probability of choosing a sample from class A: $4/10$
        - Probability of making a mistake when classifying a sample of class A: $1 - 4/10$

      - For class B:
        - Probability of choosing a sample from the class B: $3/10$
        - Probability of making a mistake when classifying a sample of class B: $1 - 3/10$

      - For class C:
        - Probability of choosing a sample from the class C : $3/10$
        - Probability of making a mistake when classifying a sample of class C: $1 - 3/10$

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Decision Trees

- CART:
  - Gini index:
    - It is calculated by adding the probability of choosing each instance multiplied for the probability of wrongly labeling it
      - For node p:

$$i(p) = \sum_{i=1}^{m} f_i(1 - f_i) = \sum_{i=1}^{m} (f_i - f_i^2) = \sum_{i=1}^{m} f_i - \sum_{i=1}^{m} f_i^2 = 1 - \sum_{i=1}^{m} f_i^2$$

      - where f_i is the ratio of instances of class i, out of m possible classes
      - It takes the value of 0 when all the instances belong the same class

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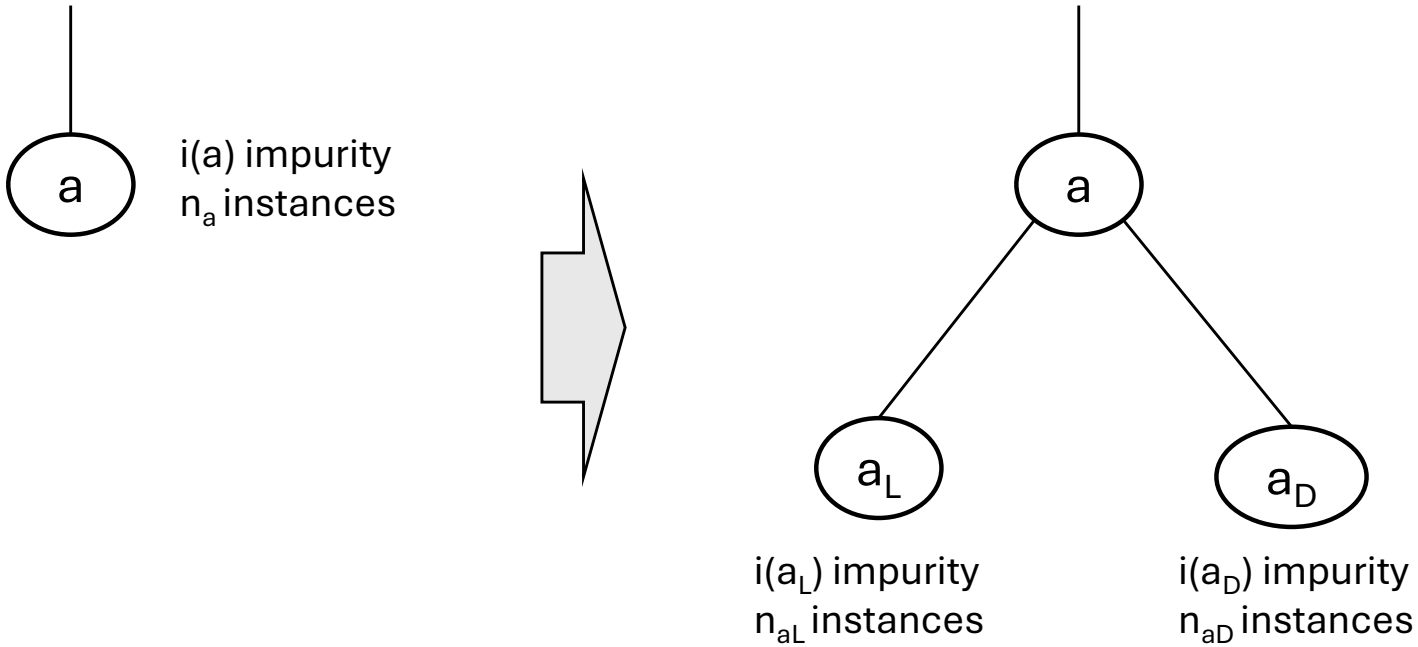
Decision Trees

- CART:
  - If the node a is divided into two parts, a_L and a_D , with a number of instances n_{aL} and n_{aD},



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Decision Trees

- CART:
  - If the node a is divided into two parts, a_L and a_D , with a number of instances n_{aL} and n_{aD}, the impurity reduction by performing this division is

$$\Delta(a) = i(a) - \frac{n_{aL}}{n_a} i(a_L) - \frac{n_{aD}}{n_a} i(a_D)$$

- The reduction of impurity is always positive
- The division that gives the maximum reduction of the impurity is chosen
  - The one that most decreases the Gini index

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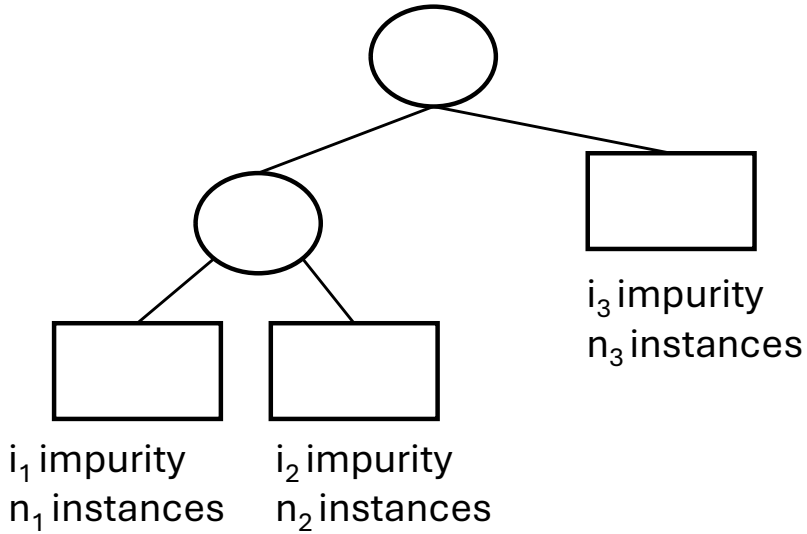
Decision Trees

- CART:
  - The impurity of a tree is the sum over all terminal nodes of the impurity of that node multiplied by the ratio of samples that reach that node in the tree
  - Example:
    - Tree A with n instances:



  - Impurity:

$$i(A) = \frac{n_1}{n} i_1 + \frac{n_2}{n} i_2 + \frac{n_3}{n} i_3$$

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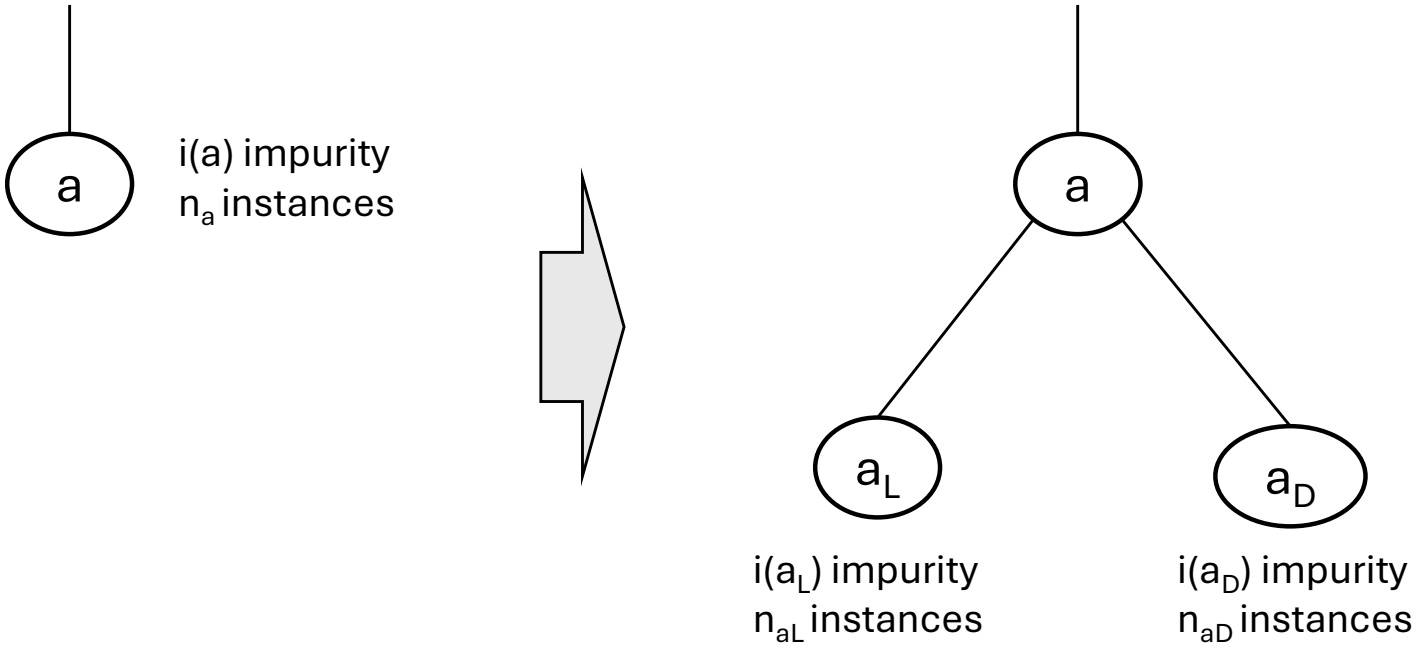
Decision Trees

- CART:
  - The impurity of a tree is the sum over all terminal nodes of the impurity of that node multiplied by the ratio of samples that reach that node in the tree
  - If we have a tree A, with a terminal node of the tree, which can be divided into two nodes a_L and a_D



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Decision Trees

- CART:
  - The impurity of a tree is the sum over all terminal nodes of the impurity of that node multiplied by the ratio of samples that reach that node in the tree
  - If we have a tree A, with a terminal node of the tree, which can be divided into two nodes a_L and a_D
    - The impurity of the new tree A' when dividing that node will be

$$i(A') = \underbrace{i(A)}_{\text{tree impurity}} + \underbrace{\frac{n_{aL}}{n} i(a_L) + \frac{n_{aD}}{n} i(a_D)}_{\text{Impurity relative to the new terminal nodes}} - \underbrace{\frac{n_a}{n} i(a)}_{\text{Impurity relative to the old terminal node a}}$$

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Decision Trees

- CART:
  - Therefore, the impurity reduction of a tree by dividing a node a would be:

$$i(A)-i(A')=\frac{n_a}{n}i(a)-\frac{n_{aL}}{n}i(a_L)-\frac{n_{aD}}{n}i(a_D)=$$

$$=\frac{n_a}{n}i(a)-\frac{n_a}{n_a}\frac{n_{aL}}{n}i(a_L)-\frac{n_a}{n_a}\frac{n_{aD}}{n}i(a_D)=$$

$$=\frac{n_a}{n}i(a)-\frac{n_a}{n}\frac{n_{aL}}{n_a}i(a_L)-\frac{n_a}{n}\frac{n_{aD}}{n_a}i(a_D)=$$

$$=\frac{n_a}{n}\left[i(a)-\frac{n_{aL}}{n_a}i(a_L)-\frac{n_{aD}}{n_a}i(a_D)\right]=\frac{n_a}{n}\Delta(a)$$

- That is, choosing the division on node a that maximizes the impurity reduction of the tree is equivalent to choosing the division that maximizes the impurity reduction of the node

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Decision Trees

- CART:
  - The growth of a tree continues until the criteria that declares a node as terminal is met:
    - If it is pure
      - There is only one class
    - If there are no possible divisions
      - For example, all instances in the node take identical values for the attributes
        - But different classes
    - A condition could be added on the minimum size of the node
      - For example, if the node contains less than 5 samples
    - An limit on the depth (maximum number of levels) of tree growth can be set

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Decision Trees

- CART:
  - The tree generated in this way correctly classifies the samples used in its learning process
    - Overfitting
    - Loss of generalization ability
    - The maximum tree is constructed from the base sample
      - Training set
  - Pruning:
    - The CART method prefers to build a very large tree in the first stage
    - In a second stage, the tree is pruned in such a way that a succession of nested subtrees is obtained
    - For this second stage, the sample-base-test is used
      - Test set (validation)

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Decision Trees

- MARS:
  - Multivariate adaptive regress Splines (Friedman, 1991)
  - Continued generalization of CART
  - Basis functions are a product of spline functions
  - A variable can be chosen multiple times
  - The tests at the nodes can be multivariate
    - With linear combinations of variables

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Decision Trees

- What problems are they suitable for?
  - Instances represented as attribute-value pairs
    - Discrete or real attributes
  - Robustness to errors in the (training) data, both in the attributes and in the response variable
  - There may be missing data for some instances in attribute values
  - Complex domains where there is no clear linear separation
  - Objective function takes discrete values (classification)
    - But it can be extended to continuous values
      - Regression trees

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Decision Trees

- Main advantages of decision trees compared to other ML techniques:
  - Easily interpretable
    - Advantage over other models such as ANNs or SVM
  - They can be executed even if a value is missing in some attribute
  - They can work with categorical inputs and/or outputs
    - No need to codify
  - They allow working with attributes with different costs
    - It is not necessary to know the value of all attributes
    - Knowing the value of high-cost attributes only if necessary
      - They will be at the bottom of the tree

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Decision Trees

- Main advantages of decision trees compared to other ML techniques:
  - Provides information on the importance of variables
    - Most important: at the top of the tree
    - Less important: at the bottom of the tree
  - Divide the search space recursively
    - They can perform complex separation regions limited by the complexity of the tree
    - They allow to work locally in each part

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